

MIT 18.06 Linear Algebra

Lecture 01. { N linear equations, N unknowns.

The Geometry of Linear Equations

Row picture

* Column picture

Matrix form

Today is July 31, 2026.

I found this manuscript in my storage room. I'm not sure when it was written. Maybe 3 years ago when I was preparing for my M.S. application.

Example 1.1 (2x2)

$$\begin{cases} 2x - y = 0 \\ -x + 2y = 3 \end{cases}$$

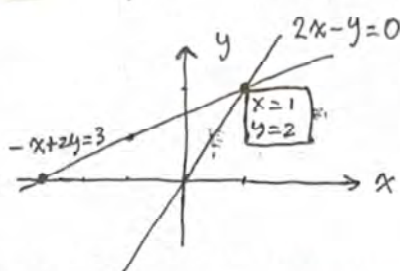
$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$A \mathbf{x} = \mathbf{b}$$

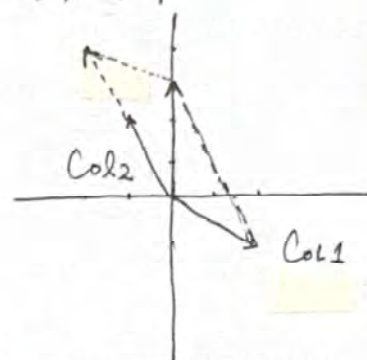
Coefficients unknowns

$$\Rightarrow x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

Row picture:



Column picture



Linear combination of columns.

Example 1.2 (3x3)

$$\begin{cases} 2x - y = 0 \\ x + 2y - z = -1 \\ -3y + 4z = 4 \end{cases}$$

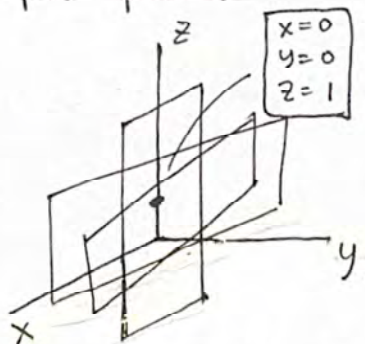
$$\begin{bmatrix} 2 & -1 & 0 \\ 1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

A

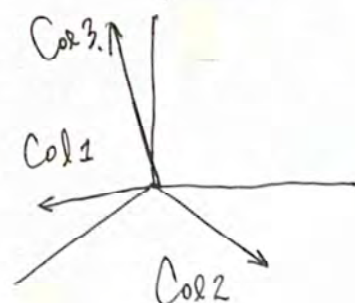
b

$$\Rightarrow x \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

Row picture



Column picture



CAN I solve $Ax=b$ for every b ?

$\Rightarrow$ Do the linear combinations of the columns fill 3-D space?

Singular Case $\rightarrow$ No.

有几维的向量对线性组合不产生贡献时。

Example 1.3 Multiple.

$$Ax=b$$

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}$$

Ax is a combination of columns of A .

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Lecture 02. Elimination with Matrices.

{ Elimination < Success
Failure

Back-Substitution (回代)

Elimination Matrices

Matrix Multiplication

Example 2.1 Elimination

$$x + 2y + z = 2.$$

$$3x + 8y + z = 12$$

$$4y + z = 2$$

$$Ax = b$$

1st pivot ($\neq 0$)

1	2	1	2
3	8	1	12
0	4	1	2

↓ ↓

A b

$$\xrightarrow{(2,1)} \begin{array}{ccc|c} \boxed{1} & 2 & 1 & 2 \\ 0 & \boxed{2} & -2 & 6 \\ 0 & 4 & 1 & 2 \end{array}$$

$$\begin{array}{ccc|c} \boxed{1} & 2 & 1 & 2 \\ \xrightarrow{(3,2)} & 0 & \boxed{2} & -2 & 6 \\ & 0 & 0 & \boxed{5} & -10 \\ & & \downarrow & & \downarrow \\ & & U & & C \end{array}$$

Back-Substitution.

$$\Rightarrow x + 2y + z = 2 \quad x = 2$$

$$2y - 2z = 6 \Rightarrow y = 1$$

$$5z = -10 \quad z = -2$$

$$U_X = C$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{matrix} 3 \times \text{Col 1} \\ + \\ 4 \times \text{Col 2} \\ + \\ 5 \times \text{Col 3} \end{matrix}$$

$$\begin{matrix} [1 & 2 & 7] \\ 1 \times 3 \end{matrix} \begin{matrix} \left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \end{array} \right] \\ 3 \times 3 \end{matrix} = \begin{matrix} 1 \times \text{Row 1} \\ + \\ 2 \times \text{Row 2} \\ + \\ 7 \times \text{Row 3} \end{matrix}$$

$$\text{Matrix} \times \text{col} = \text{col}$$

Row $\times$ Matrix = Row

Matrices:

1. Subtract $3 \times \text{Row 1}$ from Row 2.

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

$\downarrow$
 E_{21}

2. Subtract $2 \times \text{Row 2}$ from Row 3.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}$$

$\downarrow$
 E_{32}

$\otimes$

$$\boxed{E_{32}(E_{21} A) = U} \xrightarrow{\text{结合律}} \boxed{(E_{32} E_{21}) A = U}$$

Δ Permutation Matrix

Example 2.2: Exchange rows 1 & 2.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ a & b \end{bmatrix}$$

$\downarrow$
 P

左乘换行

Exchange columns 1 & 2.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} b & a \\ d & c \end{bmatrix}$$

右乘换列.

最佳思路: 不是将 $A \rightarrow U$, 而是 $U \rightarrow A$.

Inverses:

Example 2.3.

$$\begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

E_{21}^{-1}

E_{21}

I

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Lecture 03. Multiplication and Inverse Matrices

Matrix Multiplication (4 ways)
 Inverse of A AB A^T
 Gauss-Jordan / find A^{-1}

① $\begin{bmatrix} \text{---} \end{bmatrix} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} = \begin{bmatrix} \cdot \\ \cdot \\ \cdot \\ \cdot \end{bmatrix} \quad C_{34}$ $C_{34} = (\text{row 3 of } A) \cdot (\text{col 4 of } B)$
 $A^{m \times n} \quad B^{n \times p} = C^{m \times p}$
 $= a_{31}b_{14} + a_{32}b_{24} + \dots = \sum a_{3k}b_{k4}$

② $\begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} = \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}$ Columns of C are combinations of columns of B
 $A^{m \times n} \quad B^{n \times p} = C^{m \times p}$
 $A(\text{Col 1})$

③ $\begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{bmatrix} = \begin{bmatrix} \text{---} \\ \text{---} \\ \text{---} \end{bmatrix}$ Rows of C are combinations of rows of B & A ?
 $A^{m \times n} \quad B^{n \times p} = C^{m \times p}$

④ $\sum \text{column of } A \times \text{Row of } B$
 $m \times 1 \quad 1 \times p$

Example 3.1

$$\begin{bmatrix} 2 & 7 \\ 3 & 8 \\ 4 & 9 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 1 & 6 \end{bmatrix} + \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \begin{bmatrix} 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix}$$

Block Multiplication.

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} = \begin{bmatrix} \text{---} \\ \text{---} \end{bmatrix} \quad A_1B_1 + A_2B_3$$

$A \quad B$

Inverses (Square Matrices).

Adjoint Inverse

$$A^{-1}A = I, \quad AA^{-1} = I \quad \text{invertible, nonsingular.}$$

If it exists.

Singular case. No inverse.

Example 3.2.

(X=0) of A is trivial

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$$

You can find a vector X with $AX=0$

$$\begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = 0$$

Example 3.3

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

A A⁻¹ I

$$A \cdot \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}; \quad A \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

col1 col1 col2 col2.

A × Column j of A⁻¹ = column j of I.

Gauss-Jordan (Solve 2 equations at once)

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right]$$

A I

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \rightarrow \left[\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & 1 & -2 & 1 \end{array} \right]$$

I A⁻¹

$$E[A \ I] = [I \ ?]$$

$$EA = I \quad \therefore E = A^{-1} \quad \therefore EI = A^{-1}$$

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Lecture 04. Factorization into $A = LU$

$$\begin{cases} \text{Inverse of } AB, A^T \\ \text{Product of elimination matrices} \\ \underline{A = LU} \text{ (no row exchanges)} \end{cases}$$

$$AA^{-1} = I = A^{-1}A$$

$$\rightarrow (AB)(B^{-1}A^{-1}) = I, \quad (B^{-1}A^{-1})AB = I.$$

$$\rightarrow (AA^{-1})^T = I, \quad (A^{-1})^T A^T = I.$$

Example 4.1: Square (good) matrix. all pivots are fine.

$$\begin{array}{ccccc} E_{21} & A & U & A & L & U \\ \begin{bmatrix} 1 & 0 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}, & \begin{bmatrix} 2 & 1 \\ 8 & 7 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \\ & & \text{lower triangular.} & & \text{upper triangular.} \\ & & & & = \begin{bmatrix} 1 & 0 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{2} \\ 0 & 1 \end{bmatrix} \end{array}$$

$$E_{32}E_{31}E_{21}A = U$$

$$A = \underline{E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}}U$$

$$= LU$$

为什么要换过来?

(下页)

Example 4.2

$$\begin{matrix} E_{32} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -5 & 1 \end{bmatrix} \end{matrix} \begin{matrix} E_{21} \\ \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 10 & -5 & 1 \end{bmatrix} = E \quad (\text{left of } A)$$

$$\begin{matrix} E_{21}^{-1} \\ \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{matrix} \begin{matrix} E_{32}^{-1} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix} \end{matrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 5 & 1 \end{bmatrix} = L \quad (\text{left of } U).$$

If no row exchanges, multipliers go directly into L .

How many operations on $n \times n$ matrix A ?

Suppose $n=100$. (multiply, subtract.)

$$\left[\begin{array}{c} \\ \\ \\ \end{array} \right]_{100 \times 100} \longrightarrow \left[\begin{array}{ccc} \square & \dots & \\ 0 & \dots & \\ \vdots & & \\ 0 & \dots & \end{array} \right] \leftarrow \begin{array}{l} \text{About } 100^2 \text{ (除第一行, 其余元素均更新).} \\ (100 \times 99) \end{array}$$

$$\longrightarrow \left[\begin{array}{ccc} \square & \dots & \\ 0 & \square & \dots \\ \vdots & 0 & \ddots \\ 0 & 0 & \dots \end{array} \right] \leftarrow \begin{array}{l} \text{About } 99^2 \\ (99 \times 98) \end{array}$$

$\longrightarrow \dots$

$$\text{Count: } n^2 + (n-1)^2 + \dots + 1^2 \approx \frac{n^3}{3} \quad \text{on } A.$$

if b added: n^2

Permutations. $(3 \times 3) \rightarrow$ Six P 's. A group of matrix.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$I \quad P_{12} \quad P_{13} \quad P_{23}$

逆为其转.

$P^{-1} = P^T$. 互为逆

$\Rightarrow$ 延伸到 $n \rightarrow n!$

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Lecture 05. Transposes, Permutations, Space $\mathbb{R}^n$

$$\boxed{PA = LU}$$

Vector Spaces & Subspaces.

Permutations. P : execute row exchanges.

$$A = LU = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \begin{bmatrix} \diagdown & & \\ & \diagdown & \\ 0 & & \diagdown \end{bmatrix} \text{ becomes: } PA = LU$$

(for any invertible A).

P = identity matrix with reordered rows.

$n!$ counts reorderings

Counts all $n \times n$ permutations.

$$P^{-1} = P^T \quad P^T P = I$$

Example 5.1. Transpose.

$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}^T = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 3 & 1 \end{bmatrix}$$

$$(A^T)_{ij} = (A)_{ji}$$

Symmetric matrices: $A^T = A$.

Example 5.2 Symmetric.

$$\begin{bmatrix} 3 & 1 & 7 \\ 1 & 2 & 9 \\ 7 & 9 & 4 \end{bmatrix}$$

$R^T R$ is always symmetric.

Why? Take a transpose.

$$(R^T R)^T = R^T R^{TT} = R^T R$$

Chap 2.

*

Vector Spaces.

Example 5.3. $\mathbb{R}^2$ = all 2-dim real vectors. $\begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \pi \\ e \end{bmatrix}$...
= the x-y plane.

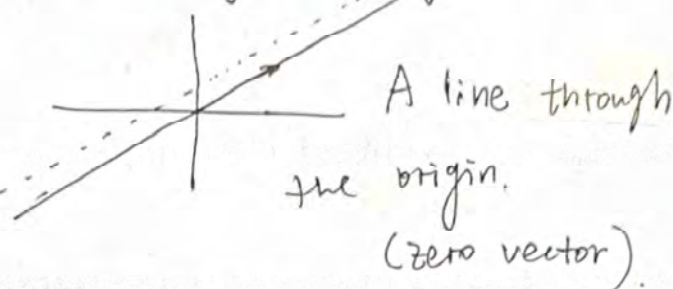
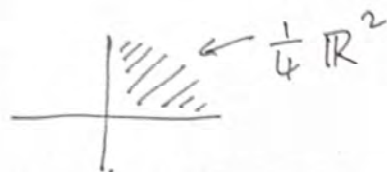
$\mathbb{R}^3$ = all vectors with 3 components.

$\vdots$ $\nwarrow$ column $\nearrow$ real.

$\mathbb{R}^n$ = all vectors with n components.

Example 5.4 not a vector space.

A vector space inside $\mathbb{R}^2$
(Subspace of $\mathbb{R}^2$).



Linear Combination is not "closed".

Subspaces in $\mathbb{R}^2$

① all of $\mathbb{R}^2$

② any line through $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$ L

③ zero vector only. Z

Example 5.5 Subspace from matrices.

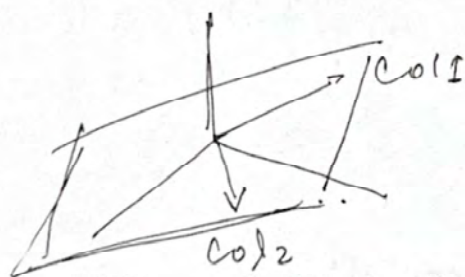
$$\begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 4 & 1 \end{bmatrix}$$

Columns in $\mathbb{R}^3$.

All their linear combinations form a subspace called "Column Space"

$C(A)$

A



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Lecture 06 Column Space and Nullspace

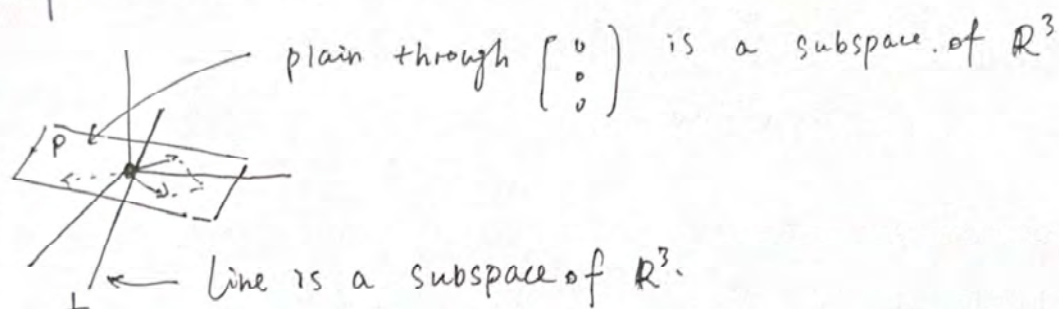
- Vector Space and Subspaces
- Column Space of A : Solving $Ax=b$
- Nullspace of A

Vector space requirements

$V+W$ and cV are in the space.

all linear combinations $cV+dW$ are in the space.

Example 6.1 $\mathbb{R}^3$



Subspaces: P and L .

$P \cup L$ = all vectors in P or L or both

This (is) (is not) a subspace of $\mathbb{R}^3$.

$P \cap L$ = all vectors in both P and L .

This (is) (is not) a subspace of $\mathbb{R}^3$.

Subspaces: S and T . intersection $S \cap T$ is a subspace.

Column space of $A_{4 \times 3}$ (is a subspace of $\mathbb{R}^4$), $C(A)$.

Example 6.2

all linear combinations of columns.

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix}$$

A

Does $Ax=b$ have a solution for every b ? NO

4 equations with 3 unknowns.

$\Downarrow$

有很多 b 向量并不在 $C(A)$ 中, 因为 A 只有三个列向量, 其线性组合不能

覆盖整个 $\mathbb{R}^4$

• Which b 's allow this system to be solved?

I can solve $Ax=b$ exactly when b is in $C(A)$.

Contains all Ax 's.

Nullspace of A = all solution $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ to the equation $Ax=0$. $N(A)$

$$Ax = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

L in $\mathbb{R}^3$.

Contains $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$\Leftrightarrow c \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

A line through $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ in $\mathbb{R}^3$

Check that solutions to $Ax=0$ always give a subspace.

If $Av=0, Aw=0$, then $\frac{A(v+w)=0}{Av+Aw}$

$$Ax = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \\ 4 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

Solutions: $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$

(Not a Subspace).

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Lecture 07. Solving $Ax=0$: Pivot Variables, Special Solutions

Computing the nullspace ($Ax=0$)

Pivot variables - free variables

Special Solutions - $\text{rref}(A)=R$

$$Ax=0 \Leftrightarrow Ux=0$$

$$\Leftrightarrow Rx=0$$

Example 7.1

$$A = \begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

2 pivot columns free columns
Echelon form (阶梯形).

rank of $A = \#$ of pivots. $= 2$.

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \rightarrow \text{Free Variable}$$

$$\begin{aligned} x_1 + 2x_2 + 2x_3 + 2x_4 &= 0 \\ 2x_3 + 4x_4 &= 0 \end{aligned}$$

$$\begin{cases} x_3 = 0 \\ x_1 = -2 \end{cases} \Rightarrow x = c \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

But it's not the whole nullspace (Special Solutions)

$$x = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 = 2 \\ x_3 = -2 \end{cases} \Rightarrow x = d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$x = c \begin{bmatrix} -2 \\ 0 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$r=2$ # pivot variables

$n-r=4-2=2$ # free variables.

Special solutions

$R =$ reduced row echelon form (zeros above + below pivots; pivots = 1)

$$\begin{bmatrix} 1 & 2 & 2 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R = \text{rref}(A)$$

Matlab

notice $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$ in pivot cols/rows.

$$I \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 2 & -2 \\ 0 & 2 \end{bmatrix} \quad F$$
 — opposite of pivot variables in special solutions.

pivot columns free columns
 0 0 0 0

⇒ proof:

Ref form $R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$
 ↑ ↑
 pivot (n-r) free
 cols cols

$x_{\text{pivot}} = -F x_{\text{free}}$

$Rx = 0 \quad \begin{bmatrix} I & F \end{bmatrix} \begin{pmatrix} x_{\text{pivot}} \\ x_{\text{free}} \end{pmatrix} = 0$

→ nullspace matrix (columns = special solution, $RN=0$)

$$N = \begin{bmatrix} -F \\ I \end{bmatrix} \quad \frac{\text{null}(A)}{\text{matlab.}}$$

Another Matrix.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 8 \\ 2 & 8 & 10 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 4 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 4 & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = U$$

$r=2$
 free col.
 (n-r=1)

Nullspace: $x = C \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} = C \begin{bmatrix} -F \\ I \end{bmatrix} = CN$

$$U \rightarrow \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix} = R$$

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Lecture 08. Solving $Ax=b$: Row Reduced Form R

$$\begin{cases} \text{Complete solution of } Ax=b \\ \text{Rank } r & X = X_p + X_n \\ r=m: \text{Solution exists} & r=n: \text{Solution is unique} \end{cases}$$

$$\begin{cases} x_1 + 2x_2 + 2x_3 + 2x_4 = b_1 \\ 2x_1 + 4x_2 + 6x_3 + 8x_4 = b_2 \\ 3x_1 + 6x_2 + 8x_3 + 10x_4 = b_3 \end{cases} \quad \begin{bmatrix} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 8 & 10 & b_3 \end{bmatrix}$$

$$[A \ b] \rightarrow \begin{bmatrix} \textcircled{1} & 2 & 2 & 2 & b_1 \\ 0 & 0 & \textcircled{2} & 4 & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & b_3 - 3b_1 \end{bmatrix} \rightarrow \begin{bmatrix} \textcircled{1} & 2 & 2 & 2 & b_1 \\ 0 & 0 & \textcircled{2} & 4 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & \underbrace{b_3 - b_2 - b_1}_0 \end{bmatrix}$$

$\uparrow$ Pivot col. $\uparrow$ $\uparrow$

Suppose $b = \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}$, $0 = b_3 - b_2 - b_1 \checkmark$ OK.

Solvability: Condition b.

$Ax=b$ is solvable when b is in $C(A)$.

• If a combination of rows of A gives zero row, then the same combination of entries of b must be zero.

To find the complete solution to $Ax=b$.

① $X_{\text{particular}}$: Set all free variables to zero. $x_2 = x_4 = 0$.

+ Solve $Ax=b$ for the pivot variables. $x_1 = -2$, $x_3 = \frac{3}{2}$

② $X_{\text{nullspace}}$.

$$Ax_p = b$$

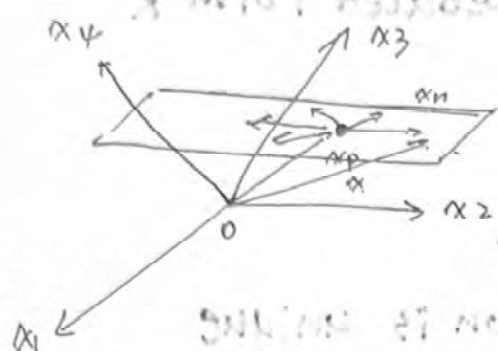
$$Ax_n = 0$$

$$\therefore X_p = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix}$$

$$X = X_p + X_n$$

$$= \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix} + c_1 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 0 \\ 0 \\ 1 \end{bmatrix} \quad A(X_p + X_n) = b$$

Plot all solutions X in $\mathbb{R}^4$



Example 8.1: $A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \\ 6 & 1 \\ 5 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$ in $b = \begin{bmatrix} 4 \\ 3 \\ 7 \\ 6 \end{bmatrix}$ case.

$\begin{bmatrix} 1 & 2 & 6 & 5 \\ 3 & 1 & 1 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & \boxed{F} \\ 0 & 1 & \boxed{F} \end{bmatrix}$

$m \times n$ matrix A of rank r . (know always $r \leq m, r \leq n$)

• Full column rank. (means $r = n$): No free variables.

$N(A) = \{ \text{zero vector} \}$

Solution $Ax = b$: $X = X_p$. — unique if it exists. (0 or 1 solutions)

• Full Row rank (means $r = m$): Can solve $Ax = b$ for every b . Exists.

Left with $\underbrace{n-r}_{n-m}$ free variables.

• $r = m = n$: 1 solution.

$A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$

$r = m = n$	$r = n < m$	$r = m < n$
$R = I$	$R = \begin{bmatrix} I \\ 0 \end{bmatrix}$	$R = \begin{bmatrix} I & 0 \end{bmatrix}$
1 solution	0/1 solution.	∞ solutions
$r < m, r < n$		

$\begin{pmatrix} 0 \\ 1 \end{pmatrix} R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$ 0/ ∞ solutions.

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Lecture 09. Independence, Basis, and Dimension

Linear Independence
Spanning a space
BASIS and dimension

Suppose A is $m \times n$ with $m < n$. Then, there're nonzero solutions to $Ax = 0$.

(more unknowns
than equations).

$$\begin{bmatrix} 0 & 0 \\ \vdots & \vdots \\ 0 & 0 \end{bmatrix}$$

Reason: There'll be free variables.

Background ✖

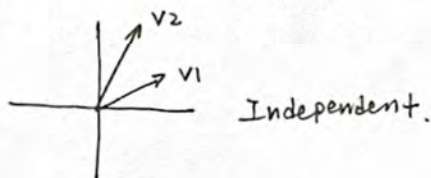
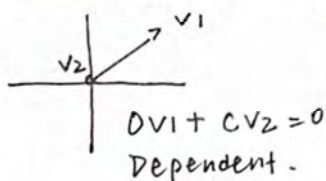
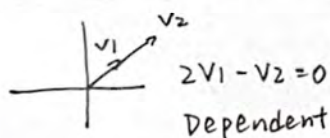
Independence

Vectors $x_1, x_2, \dots, x_n$ are independent if ...?

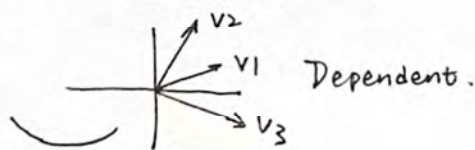
no combination gives the zero vector (except the zero combination)
all $c_i = 0$.

$$c_1 x_1 + c_2 x_2 + \dots + c_n x_n \neq 0$$

Example 9.1



$$A = \begin{bmatrix} v_1 & v_2 & v_3 \\ 2 & 1 & 2.5 \\ 1 & 2 & -1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



Repeat: When $v_1, \dots, v_n$ are columns of A .

They are independent if $N(A)$ is {zero vector}.

They are dependent if $A \cdot c = 0$ for some non-zero c .

no free variables
 $\text{rank} = n, N(A) = \{0\}$

$\text{rank} < n$.
yes free variables

Vectors $v_1, \dots, v_d$ span a space means the space consists of all combinations of those vectors.

Basis for a space is a sequence of vectors $v_1, v_2, \dots, v_d$ with 2 properties

1. They're independent.

2. They span the space.

Example 9.2

Space is $\mathbb{R}^3$, one basis is $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ — standard basis.

Another basis is $\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \\ 8 \end{bmatrix}$

$\mathbb{R}^n$ n vector give basis if the $n \times n$ matrix with those columns is invertible.

Given a space. Every basis for the space has the same number of vectors.

Def the dimension of the space.

Example 9.3.

Space is $C(A)$.

$$\begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

↑ ↑ ↑ ↑

$$N(A) \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\dim N(A) = \# \text{ free variables} = n - r.$$

$$\text{rank}(A) = \# \text{ pivot columns} = \underbrace{\dim C(A)}_{\text{space}} = 2.$$

(matrix) (space)

Basis: Col 1, 2 / Col 1, 3 / Col 2, 4 / ... $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} / \dots$

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Lecture 10. The Four Fundamental Subspaces

• Four Fundamental Subspaces (for Matrix A)

4 Subspaces. $A_{m \times n}$.

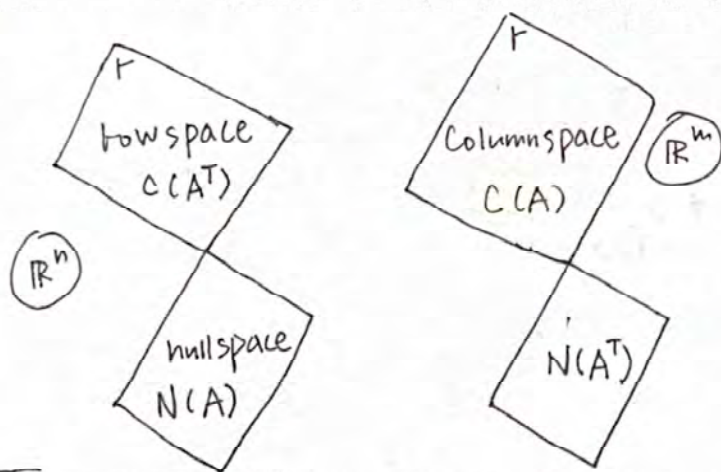
$$\begin{bmatrix} A \\ n. \end{bmatrix}_m \begin{bmatrix} \\ 1. \end{bmatrix}_n$$

• Columnspace = $C(A)$ in $\mathbb{R}^m$

• Nullspace = $N(A)$ in $\mathbb{R}^n$

• Rowspace = all combinations of rows of A = all combinations of columns of A^T .
 = $C(A^T)$ in $\mathbb{R}^n$

• Nullspace of $A^T = N(A^T)$ = Left nullspace of A.
 in $\mathbb{R}^m$



basis?

dimension?

	basis	dimension
$C(A)$	pivot column	r
$C(A^T)$	first r rows of R	r
$N(A)$	special solutions	$n-r$
$N(A^T)$		$m-r$

Example 10.1

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R.$$

Because of row exchanges,

$$C(A) \neq C(R).$$

$$\text{But } C(A^T) = C(R^T)$$

and their basis is first r rows of R .

4th space: $N(A^T)$ $A^T y = 0$

$$A^T y = 0$$

$$y^T A^T = 0^T$$

$$y^T A = 0^T$$

$$\begin{bmatrix} y^T \\ \text{left} \end{bmatrix} \begin{bmatrix} A \\ \end{bmatrix} = \begin{bmatrix} 0^T \\ \end{bmatrix}$$

$$[A_{m \times n} \ I_{m \times m}] \xrightarrow{\text{ref}} [R_{m \times n} \ E_{m \times m}]$$

$$E(A) \xrightarrow{\text{ref}} [R_{m \times n} \ E_{m \times m}]$$

$EA = R$ in Chapter 2, R was I . Then E was A^{-1} .

Example 10.2

$$[A \ I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 \\ 1 & 2 & 3 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & -1 & 2 & 0 \\ 0 & 1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 \end{array} \right] = [R \ E]$$

$R = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

who produced 0 how?

$E = \begin{bmatrix} -1 & 2 & 0 \\ 1 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$

$N(AT)$

New vector space M .

All 3×3 matrices.

$A+B, cA$ (not AB for now)

Subspaces of M : upper triangular matrices,
symmetric matrices

diagonal matrices D $\dim D = 3$
 $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

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Lecture 11. Matrix Spaces; Rank 1; Small world Graphs

{ Bases of new vector spaces $\rightarrow M =$ all 3×3 matrices.
Rank 1 matrices $\xrightarrow{\text{sub}}$ Symmetric 3×3 $\dim S = 6$.
Small world Graphs upper triangular 3×3 $\dim U = 6$

Basis for $M =$ all 3×3 's. $\dim M = 9$.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$S \cap U =$ symmetric and uppertriangular. = diagonal 3×3 's. $\dim(S \cap U) = 3$.

$S \cup U$ is not a subspace.

$\rightarrow S + U =$ any element of S + any element of U . = all 3×3 's.

$$\dim(S + U) = 9.$$

$$\boxed{\dim S + \dim U = \dim(S \cap U) + \dim(S + U)}$$

Example 11.1

$$\frac{d^2 y}{dx^2} + y = 0$$

BASIS.

$$\Rightarrow y = \cos x, \sin x, (e^{ix}).$$

$$\Rightarrow \text{complete solution: } y = c_1 \cos x + c_2 \sin x.$$

$$\dim(\text{solution space}) = 2.$$

Rank 1. — Building Block.

$$\dim C(A) = \text{rank } A = \dim C(A^T).$$

$$A_{2 \times 3} = \begin{bmatrix} 1 & 4 & 5 \\ 2 & 8 & 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 4 & 5 \end{bmatrix} - \text{rank 1 matrix } A = uv^T$$

$$r = 1.$$

Example 11.2.

Linear Algebra

$M =$ all 5×7 matrices.

Subset of rank 4 matrices is not a subspace.

Rank 4 matrix $X_1 +$ rank 4 matrix $X_2 =$ rank 5 matrix.

Subset of rank 1 matrices — not a subspace.

Example 11.3.

In $\mathbb{R}^4$. $V = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix}$ $S =$ all v in $\mathbb{R}^4$ with $v_1 + v_2 + v_3 + v_4 = 0$.

$=$ nullspace of $A = [1 \ 1 \ 1 \ 1]$ $r(A) = 1$, $\dim N(A) = n - r = 3$.

$AV = [1 \ 1 \ 1 \ 1] \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = 0$.

Basis for $S = \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$

$C(A) = \mathbb{R}^1$ $\dim C(A) = 1$.

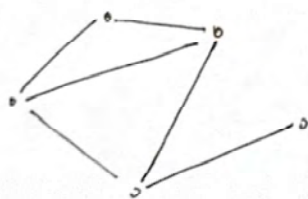
$N(A^T) = \{0\}$ $\dim N(A^T) = 0$.

$C(A^T) = C \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ $\dim C(A^T) = 1$.

$\dim N(A) + \dim C(A^T) = 3 + 1 = 4$.

$\dim N(A^T) + \dim C(A) = 0 + 1 = 1$.

Graph: {nodes, edges}

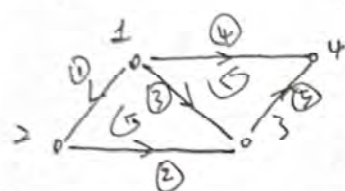


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Lecture 12. Graphs, Networks, Incidence Matrices

- Graphs + Networks
- Incidence Matrices
- Kirchhoff's Laws

Graph: Nodes, Edges.



$n=4$ nodes
 $m=5$ edges

Incidence matrix A :

	node 1	2	3	4	
edge 1	-1	1	0	0	} loop
2	0	-1	1	0	
3	-1	0	1	0	
4	-1	0	0	1	
5	0	0	-1	1	

$$Ax=0 \quad \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_2 - x_1 \\ x_3 - x_2 \\ x_3 - x_1 \\ x_4 - x_1 \\ x_4 - x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

↓
Potentials of the nodes → Potentials differences.

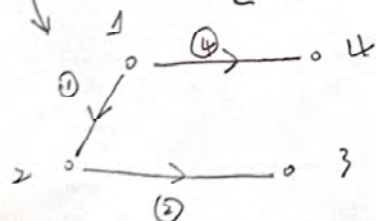
$x = c \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$
 $\dim N(A) = 1$
 $\text{rank } A = 3$

Kirchhoff's CL:

$$A^T y = 0 \quad \begin{bmatrix} -1 & 0 & -1 & -1 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \\ y_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

↓
OHM's Law

$y = c_1 \begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 0 \\ 0 \\ -1 \\ 1 \end{bmatrix}$
 $\dim N(A^T) = m - r = 2$



Tree: no loops.

(rank = $n-1$)

$$\dim N(A^T) = m - r \quad \boxed{\# \text{ loops} = \# \text{ edges} - (\# \text{ nodes} - 1)}$$

$$\# \text{ nodes} - \# \text{ edges} + \# \text{ loops} = 1$$

Euler's formula.

$$X = x_1, x_2, x_3, x_4$$

Potentials at nodes

$$\downarrow E = Ax.$$

$x_i - x_j$, etc.

Potential differences

$$\xrightarrow[\text{OHM's Law}]{y = ce}$$

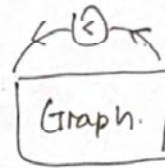
$y = y_1, y_2, y_3, y_4$
Currents on edges.

$A^T y = 0$ Kirchhoff's CL.

$$\uparrow A^T$$

$\downarrow$ Sources in.

$$A^T y = f.$$



$$A^T C A x = f$$

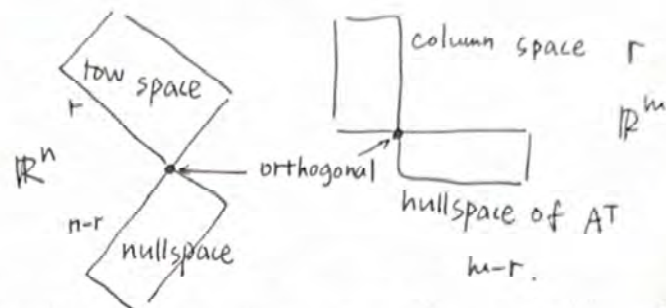
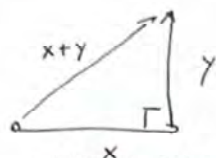
$\downarrow$
Symmetric

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Lecture 14. Orthogonal Vectors and Subspaces

Orthogonal vectors + subspaces
 nullspace $\perp$ row space
 $N(A^T A) = N(A)$

Orthogonal vectors.



Pythagoras.

$$x^T y = x_1 y_1 + x_2 y_2 + \dots = 0 \quad (\text{dot product})$$

inner product

→ proof: $\|x\|^2 + \|y\|^2 = \|x+y\|^2$

$$\begin{matrix} \downarrow & \downarrow & \downarrow \\ x^T x + y^T y & = & (x+y)^T (x+y) = x^T x + y^T y + \underbrace{x^T y + y^T x}_0 \end{matrix}$$

$x^T y = y^T x$

Example 14.1. $x = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $y = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$, $x+y = \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}$

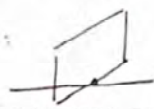
$$\|x\|^2 = 14, \quad \|y\|^2 = 5, \quad \|x+y\|^2 = 19.$$

$$\therefore \|x\|^2 + \|y\|^2 = \|x+y\|^2 \therefore x \text{ and } y \text{ are orthogonal.}$$

Subspace S is orthogonal to subspace T.

Means: every vector in S is orthogonal to every vector in T.

Example 14.2:



Never.



Always



Row space is orthogonal to nullspace. WHY?

$$Ax = 0$$

$$\begin{bmatrix} \text{row 1} \\ \text{row 2} \\ \vdots \\ \text{row } m \end{bmatrix} \begin{bmatrix} x \end{bmatrix} = 0$$

$$\Rightarrow \begin{cases} (\text{row 1}) \cdot x = 0 \\ (\text{row 2}) \cdot x = 0 \\ \vdots \\ (\text{row } m) \cdot x = 0 \end{cases} \Rightarrow (c_1 \text{row 1} + c_2 \text{row 2} + \dots) x = 0$$

Example 14.3

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 2 & 4 & 10 \end{bmatrix}$$

$$n=3, r=1.$$

$$\dim N(A) = 2, \dim C(A^T) = 1.$$

Nullspace and row space are orthogonal complements in $\mathbb{R}^n$.
正交补.

Nullspace contains all vectors $\perp$ row space.

Coming: $Ax = b$. "Solve" it when there is no solution.

rectangular ($m > n$) - get noise in right hand side. no solution.

$$\underline{A^T A}_{n \times n}$$

$$\text{symmetric: } (A^T A)^T = A^T A^{TT} = A^T A.$$

$$A^T A \hat{x} = A^T b$$

"good" equation.

Example 14.4.

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

no solution

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 5 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\text{invertible} - \begin{bmatrix} 3 & 8 \\ 8 & 30 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} b_1 + b_2 + b_3 \\ b_1 + 2b_2 + 5b_3 \end{bmatrix}$$

But:

$$\begin{bmatrix} 1 & 1 & 1 \\ 3 & 3 & 3 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 1 & 3 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 9 \\ 9 & 27 \end{bmatrix} \sim \text{not invertible.}$$

rank of $A^T A$ = rank of A . ✓

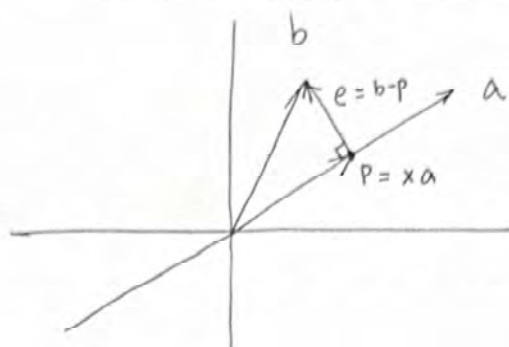
$$N(A^T A) = N(A)$$

$\Rightarrow A^T A$ is invertible exactly if A has independent columns.

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Lecture 15. Projections onto Subspaces

Projections!
Least Squares
Projection Matrix



$$a^T(b - xa) = 0$$

$$\Rightarrow x = \frac{a^T b}{a^T a}$$

$$p = ax = xa$$

$$= a \cdot \frac{a^T b}{a^T a}$$

$$= P b$$

projection $p = P b$
 $\downarrow$
 projection matrix

$$P = \frac{a a^T}{a^T a} \rightarrow \#$$

$C(P) =$ a line through A .

$$\text{rank}(P) = 1$$

$$\begin{cases} * P^T = P \\ * P^2 = P \end{cases} \text{ - symmetric}$$

Why project?

Because $Ax = b$ may have no solution.

Solve $A\hat{x} = p$ instead.

$\hookrightarrow$ projection of b onto $C(A)$.

$$p = \hat{x}_1 a_1 + \hat{x}_2 a_2 = A\hat{x} \quad \text{Find } \hat{x}.$$

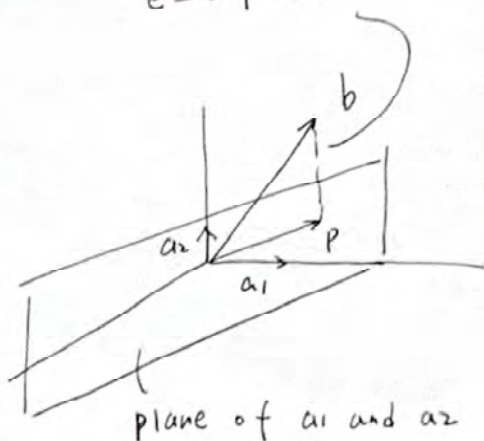
Key: $\underbrace{b - A\hat{x}}_e$ is perp. to plane.

$$\Rightarrow \begin{cases} a_1^T (b - A\hat{x}) = 0 \\ a_2^T (b - A\hat{x}) = 0 \end{cases}$$

$$\begin{bmatrix} -a_1^T \\ -a_2^T \end{bmatrix} (b - A\hat{x}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \rightarrow A^T (b - A\hat{x}) = 0.$$

$$\rightarrow A^T A \hat{x} = A^T b \rightarrow \hat{x} = (A^T A)^{-1} A^T b. \quad (\text{NEXT PAGE})$$

$e = b - p$ is perp. to plane



in $N(A^T) \Rightarrow e \perp C(A)$ = col space of $A = \begin{bmatrix} a_1 & a_2 \end{bmatrix}$

$$\hat{x} = (A^T A)^{-1} A^T b$$

$$P = A \hat{x} = A (A^T A)^{-1} A^T b$$

ID case $\frac{a a^T}{a^T a}$

$$P = A (A^T A)^{-1} A^T$$

$\hookrightarrow$ if A is a square matrix, then $A^{-1} (A^{-1})^T$ (invertible)

$$P = A A^{-1} (A^{-1})^T A^T = I.$$

Two properties: $P^T = P$, $P^2 = P$.

$\downarrow$

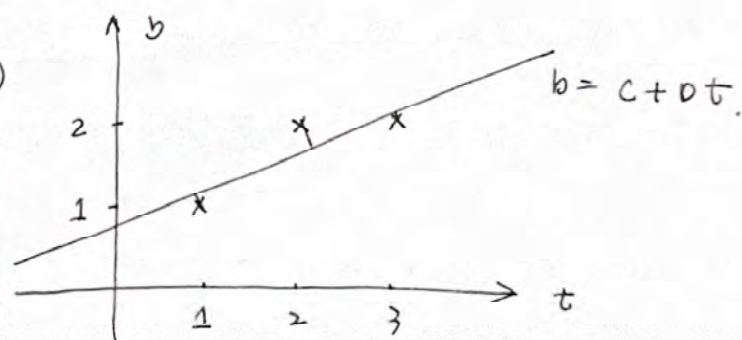
$$\begin{aligned} P^2 &= A (A^T A)^{-1} A^T A (A^T A)^{-1} A^T \\ &= A (A^T A)^{-1} (A^T A (A^T A)^{-1}) A^T \\ &= A (A^T A)^{-1} A^T = P \end{aligned}$$

Least square Fitting by a line.

(1, 1), (2, 2), (3, 2)

$$\begin{cases} C + D = 1 \\ C + 2D = 2 \\ C + 3D = 2 \end{cases} \quad \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

$$A x = b$$



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Lecture 16. Projection Matrices and Least Squares.

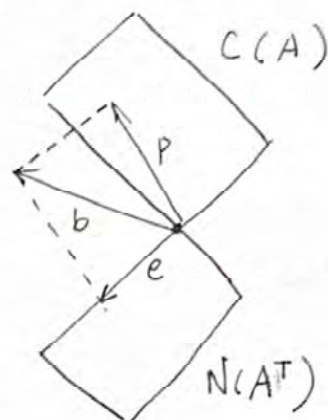
Projections

Least squares and best straight line.

Proj-Matrix $P = A(A^T A)^{-1} A^T$

If b in column space $Pb = b$

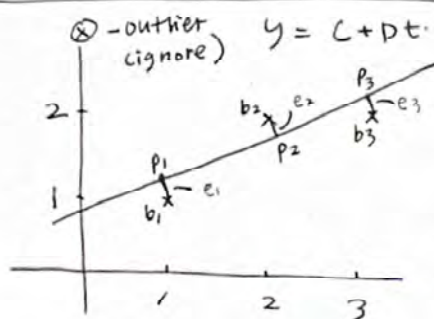
If $b \perp$ column space $Pb = 0$



$$b = P + e = Pb + \underbrace{(I-P)b}_{\substack{\downarrow \\ \text{Proj. onto } \perp \\ \text{space}}}$$

Example 16.1.

$$\begin{cases} C + D = 1 \\ C + 2D = 2 \\ C + 3D = 2 \end{cases} \quad Ax = b$$



$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Minimize $\|Ax - b\|^2 = \|e\|^2 = e_1^2 + e_2^2 + e_3^2$

$$\begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix} = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}$$

Find p . $\hat{x} = \begin{bmatrix} \hat{C} \\ \hat{D} \end{bmatrix}$:

① $A^T A \hat{x} = A^T b$ — normal equations.

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 6 & 5 \\ 6 & 14 & 11 \end{bmatrix}$$

$$\begin{cases} 3\hat{C} + 6\hat{D} = 5 \\ 6\hat{C} + 14\hat{D} = 11 \end{cases}$$

$$\rightarrow \begin{cases} \hat{C} = \frac{2}{3} \\ \hat{D} = \frac{1}{2} \end{cases}$$

→ Linear Algebra

② $\|e\|^2 = (\hat{C} + \hat{D} - 1)^2 + (\hat{C} + 2\hat{D} - 2)^2 + (\hat{C} + 3\hat{D} - 2)^2$

$$\frac{\partial^2 \|e\|^2}{\partial \hat{C}^2} = 3\hat{C} + 6\hat{D} - 5 = 0$$

$$\rightarrow \begin{cases} \hat{C} = \frac{2}{3} \\ \hat{D} = \frac{1}{2} \end{cases}$$

$$\frac{\partial^2 \|e\|^2}{\partial \hat{D}^2} = 6\hat{C} + 14\hat{D} - 11 = 0$$

→ Calculus.

∴ Best line: $y = \frac{2}{3} + \frac{1}{2}t$. Error vector $e = \begin{bmatrix} \frac{1}{3} \\ -\frac{1}{3} \\ \frac{1}{6} \end{bmatrix}$ ($\perp$ to all vectors in $C(A)$.)

$$\underline{A^T A}$$

orthogonal matrix

If A has independent columns, then $A^T A$ is invertible.

→ proof: Suppose $A^T A x = 0$, x must be zero vector.

IDEA: $x^T A^T A x = 0$.

and $x^T A^T A x = 0$ has zero vector.

→ $\underbrace{(Ax)^T (Ax)}_{\text{length of } Ax} = 0$.

⇒ $Ax = 0$.

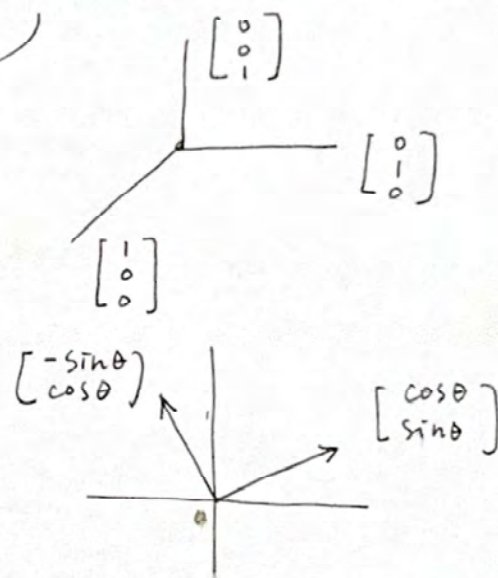
independent ⇒ $x = 0$.

✗

Columns definitely independent if they're perpendicular unit vectors.

orthonormal
vectors.

标准正交向量.



MIT 18.06 Linear Algebra

Lecture 17. Orthogonal Matrices and Gram-Schmidt

- Orthogonal basis $q_1, q_2, \dots, q_n$
- Orthogonal matrix Q : square.
- Gram-Schmidt $A \rightarrow Q$

Orthogonal vectors: $q_i^T q_j = \begin{cases} 0, & \text{if } i \neq j \\ 1, & \text{if } i = j \end{cases}$

$$Q = \begin{bmatrix} | & & | \\ q_1 & \dots & q_n \\ | & & | \end{bmatrix} \quad Q^T Q = \begin{bmatrix} - & q_1^T & - \\ \vdots & \vdots & \vdots \\ - & q_n^T & - \end{bmatrix} \begin{bmatrix} | & & | \\ q_1 & \dots & q_n \\ | & & | \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} = I$$

If Q is square, then $Q^T Q$ tells us $Q^T = Q^{-1}$

Example 17.1

Permutation: $Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \quad Q^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} \quad Q^T Q = I$

Example 17.2

$$Q = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

Example 17.3

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Example 17.4

$$Q = \frac{1}{2} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \quad \text{Adhemar Matrix.}$$

Example 17.5 Rectangular Case

$$Q = \frac{1}{3} \begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}$$

Q has orthonormal columns. Project onto its column space.

$$P = Q(Q^T Q)^{-1} Q^T = Q Q^T = \begin{cases} I & \text{if } Q \text{ is square} \end{cases} \quad \left\{ \begin{array}{l} (Q Q^T)(Q Q^T) = Q Q^T \\ (Q Q^T)^T = Q^T Q = Q Q^T \end{array} \right. \quad \begin{array}{l} \text{properties} \\ \text{checked,} \end{array}$$

$$A^T A \hat{x} = A^T b \quad \text{Now: } A^T A \hat{x} = A^T b$$

② only square case? $\Rightarrow \quad \hat{x} = A^T b, \quad \hat{x}_i = q_i^T b.$

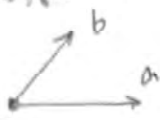
Gram - Schmidt.

orthogonal matrices

Orthogonal - more than 2D

Vectors a, b .

They're independent.



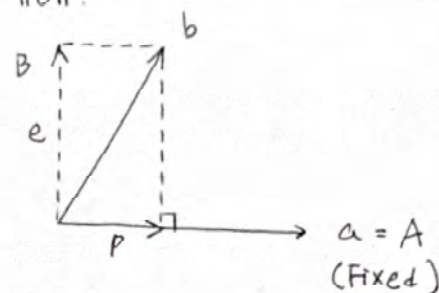
Goal: orthogonal $A, B \rightarrow$ orthonormal vectors.

$$q_1 = \frac{A}{\|A\|}$$

$$q_2 = \frac{B}{\|B\|}$$

$$\bullet \quad \underbrace{A = a}_{\text{Fixed}} \quad B = b - \frac{A^T b}{A^T A} A$$

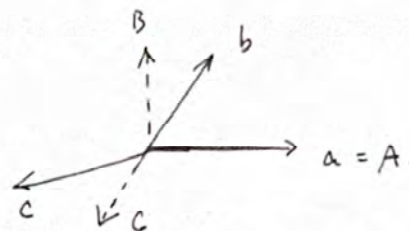
$$\text{Check } A \perp B: A^T B = A^T (b - \frac{A^T b}{A^T A} A) = 0 \quad \checkmark$$



Vectors a, b, c .

$$A = a \quad B = b - \frac{A^T b}{A^T A} A \quad C = c - \frac{A^T c}{A^T A} A - \frac{B^T c}{B^T B} B$$

$$(C \perp A, C \perp B)$$



$$q_1 = \frac{A}{\|A\|} \quad q_2 = \frac{B}{\|B\|} \quad q_3 = \frac{C}{\|C\|}$$

Example 7.6.

$$a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$A = a \quad B = b - \frac{A^T b}{A^T A} A = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} - \frac{3}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$Q = [q_1 \ q_2] = \begin{bmatrix} 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}$$

Elimination: $A = LU$

Orthogonal: $A = QR$

$$A = [a_1 \ a_2] = [q_1 \ q_2] R = [q_1 \ q_2] \begin{bmatrix} \nabla & \\ 0 & \end{bmatrix} = [q_1 \ q_2] \begin{bmatrix} a_1^T q_1 & a_2^T q_1 \\ a_1^T q_2 & a_2^T q_2 \end{bmatrix}$$

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Lecture 18. Properties of Determinants

$$\begin{cases} \text{Determinants } \det A = |A| \\ \text{Properties 1.2.3.4-10} \\ + \text{ signs} \end{cases}$$

Properties.

1. $\det I = 1$

2. Exchange rows reverse the sign of the determinant. $\left. \begin{array}{l} \text{ } \end{array} \right\} \det P = \begin{matrix} 1 & -1 \\ \text{even} & \text{odd} \end{matrix}$

① $\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1$ ② $\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$

3. a. $\begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

b. $\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix}$

Linear for each row.

4. 2 equal rows $\rightarrow \det = 0$.

$\rightarrow$ proof: Exchange those rows $\rightarrow$ same matrix. $\left. \begin{array}{l} \text{prop} \rightarrow \text{reverse the sign} \end{array} \right\} \Rightarrow \det = 0$.

5. Subtract $l \times \text{row } i$ from row k . The det doesn't change.

$$\begin{vmatrix} a & b \\ c-la & d-lb \end{vmatrix} \stackrel{③}{=} \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a & b \\ -la & -lb \end{vmatrix} \stackrel{③}{=} \begin{vmatrix} a & b \\ c & d \end{vmatrix} - l \begin{vmatrix} a & b \\ a & b \end{vmatrix} \stackrel{④}{=} \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

6. Row of zeros $\rightarrow \det = 0$.

$$0 \begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a \cdot 0 & b \cdot 0 \\ c & d \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ c & d \end{vmatrix} \Rightarrow \det = 0.$$

7. $U = \begin{bmatrix} d_1 & & \\ & d_2 & \\ & & \ddots \\ 0 & & & d_n \end{bmatrix}$ (upper triangular) $\Rightarrow \det U = d_1 d_2 \cdots d_n$ (product of the pivots)

8. $\det A = 0$ when A is singular.
 $\det A \neq 0$ when A is invertible.

$$9. \det AB = (\det A)(\det B)$$

$$\det A^{-1} = \frac{1}{\det A}$$

$$\det tA = t^n \det A$$

$$10. \det A^T = \det A$$

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & c \\ b & d \end{vmatrix}$$

$$\rightarrow \text{proof: } \begin{vmatrix} A \end{vmatrix} \stackrel{①}{=} |L||U| \quad \Rightarrow \quad |L||U| \stackrel{①}{=} |U^T||L^T|$$

$$\begin{vmatrix} A^T \end{vmatrix} \stackrel{②}{=} |U^T||L^T|$$

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Lecture 19. Determinant Formulas and Cofactors

- Formula for $\det A$ ($n!$ terms)
- Cofactor formula
- Tridiagonal matrices

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix} = \underbrace{\begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix}}_{0} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \underbrace{\begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix}}_{0} + \underbrace{\begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix}}_{0}$$

$$= ad - bc$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{vmatrix} + \begin{vmatrix} a_{11} & 0 & 0 \\ 0 & 0 & a_{23} \\ 0 & a_{32} & 0 \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ a_{21} & 0 & 0 \\ 0 & 0 & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ 0 & 0 & a_{23} \\ a_{31} & 0 & 0 \end{vmatrix} + \square + \square$$

$$= a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}$$

Big Formula. $n \times n \det.$

$$\det A = \sum_{n! \text{ terms}} \pm a_{1\alpha} a_{2\beta} a_{3\gamma} \dots a_{nw}, \quad (\alpha, \beta, \gamma, \dots, w) = \text{perm of } (1, 2, 3, \dots, n)$$

Example 19.1

$$\begin{vmatrix} 0 & 0 & \textcircled{1} & 1 \\ 0 & \textcircled{1} & 1 & 0 \\ \textcircled{1} & 1 & 0 & 0 \\ 1 & 0 & 0 & \textcircled{1} \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{vmatrix} \xrightarrow{(4,3,2,1) \rightarrow +} \begin{vmatrix} 3 & 2 & 1 & 4 \end{vmatrix} \rightarrow -1 = 0.$$

Cofactors 3×3

$$\det = a_{11} (a_{22}a_{33} - a_{23}a_{32}) + a_{12} (a_{23}a_{31} - a_{21}a_{33}) + a_{13} (a_{21}a_{32} - a_{22}a_{31})$$

$$= \begin{vmatrix} a_{11} & & \\ & a_{22} & a_{23} \\ & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} & a_{12} & \\ a_{21} & & \\ a_{31} & & \end{vmatrix} + \begin{vmatrix} & & a_{13} \\ a_{21} & a_{22} & \\ a_{31} & a_{32} & \end{vmatrix}$$

→ Cofactor of $a_{ij} = \pm \det$ (n-1 matrix without row i erased col j)

(C_{ij})

$$\begin{cases} + & \text{if } i+j \text{ is even} \\ - & \text{if } i+j \text{ is odd.} \end{cases}$$

Cofactor Formula (along row 1)

$$\det A = a_{11} C_{11} + a_{12} C_{12} + \dots + a_{1n} C_{1n}$$

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = a \cdot |d| + b(-|c|) = ad - bc$$

Example 19.2

$$\begin{vmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{vmatrix}$$

A_n

$$|A_1| = 1 \quad |A_2| = 0 \quad |A_3| = -1$$

$$|A_4| = 1 \cdot |A_3| - 1 \cdot |A_2| = -1 \quad |A_5| = 0 \quad |A_6| = 1 \quad |A_7| = 1 \dots$$

$$\downarrow$$

$$|A_n| = |A_{n-1}| - |A_{n-2}|$$

sequence:

$$1 \quad 0 \quad -1 \quad -1 \quad 0 \quad 1 \quad 1 \dots$$

period 6.

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Lecture 20. Cramer's Rule, Inverse Matrix, and Volume

- Formula for A^{-1}
- Cramer's Rule for $x = A^{-1}b$
- $|\det A| = \text{Volume of box}$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det A} C^T \rightarrow \text{products of } n-1 \text{ entries.}$$

→ products of n entries

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}^{-1}$$

Check $AC^T = (\det A)I$

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} c_{11} & \dots & c_{n1} \\ \vdots & \ddots & \vdots \\ c_{1n} & \dots & c_{nn} \end{bmatrix} = \begin{bmatrix} \det A & & \\ & \det A & \\ & & \ddots \\ & & & \det A \end{bmatrix}$$

WHY:
Look at
the book

$$Ax = b \quad x = A^{-1}b = \frac{1}{\det A} C^T b.$$

Cramer's Rule:

$$x_1 = \frac{\det B_1}{\det A}$$

$$B_1 = \begin{bmatrix} | & n-1 \text{ cols} \\ b & \text{of } A \end{bmatrix} = \text{A with column 1 replaced by } b.$$

$$x_2 = \frac{\det B_2}{\det A}$$

↓

↳ proof: Look at the book.

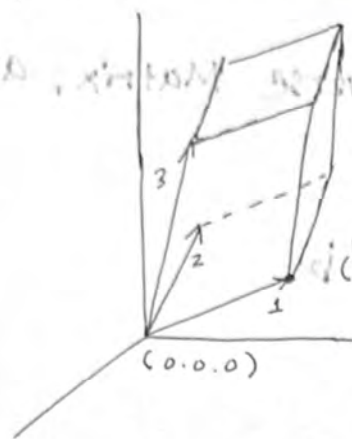
⋮

$$x_j = \frac{\det B_j}{\det A}$$

$$B_j = \text{A with column } j \text{ replaced by } b.$$

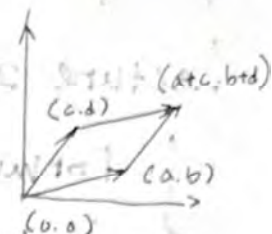
$|\det A| = \text{volume of box.}$ $\longrightarrow$ properties 1. 2. 3 a 3 b

3x3.



$$\text{area} = \det \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\rightarrow ad - bc$$



special case: $A = I$ Identity cube (unit cube)

orthogonal case: $A = Q$.

$\uparrow$
rotate

$$Q^T Q = I$$

$$|Q^T| |Q| = |I|$$

$$|Q|^2 = 1$$

$$Q = \pm 1$$

$\downarrow$

Lecturer didn't do that well. Look at the book. by yourself.

area

$$= \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

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Lecture 21. Eigenvalues and Eigenvectors

$$\begin{cases} \text{Eigenvalues - Eigenvectors} \\ \det[A - \lambda I] = 0 \\ \text{Trace} = \lambda_1 + \lambda_2 + \dots + \lambda_n \end{cases}$$

Ax parallel to x - Eigenvectors

$$Ax = \lambda x$$

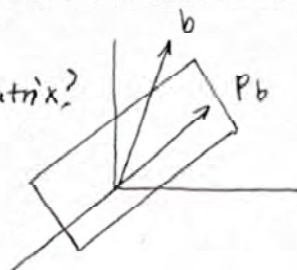
If A is singular, then $\lambda = 0$ is an eigenvalue.

Example 21.1

What are x 's and λ 's for projection matrix?

→ Any x in the plane: $Px = x$. ($\lambda = 1$)

→ Any $x \perp$ plane: $Px = 0$. ($\lambda = 0$)



Example 21.2

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \lambda = 1; \quad x = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, \lambda = -1.$$

$$Ax = x$$

$$Ax = -x$$

Fact: Sum of λ 's = $a_{11} + a_{22} + \dots + a_{nn}$. — Trace.

How to solve $Ax = \lambda x$.

Rewrite: $(A - \lambda I)x = 0$
Singular.

↓
→ $|A - \lambda I| = 0$ Find λ first.

Example 21.3.

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = (3-\lambda)^2 - 1 = 0$$

$$\lambda^2 - 6\lambda + 8 = 0$$

$$\begin{matrix} \Delta \\ \downarrow \\ \text{Trace} \end{matrix} \quad \begin{matrix} \downarrow \\ \det(A) \end{matrix}$$

$$\lambda_1 = 4, \lambda_2 = 2 \quad \text{or} \quad [\text{Tr}(A) - \lambda] = 0$$

$$\lambda_1 + \dots + \lambda_n + 1\lambda = \text{Trace}$$

$$A - 4I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \quad x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix};$$

$$A - 2I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \star$$

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\lambda_1 = 1, x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \quad \lambda_2 = -1, x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

+
3I
=

$$\text{If } Ax = \lambda x : (A + 3I)x = \lambda x + 3x = (\lambda + 3)x$$

$$\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\lambda_1 = 4, x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \quad \lambda_2 = 2, x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Generalize.

If $Ax = \lambda x$, B has eigenvalues α , then $(A+B)x = (\lambda + \alpha)x$
 ~~$(A+B)x = (\lambda + \alpha)x$~~
 $(Bx \neq \lambda x)$

So not so great $A+B$, AB .

Example 21.4. 90° rotation.

$$Q = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\text{trace} = 0 + 0 = 0 = \lambda_1 + \lambda_2$$

$$\det Q = 1 = \lambda_1 \lambda_2$$

$$\det(Q - \lambda I) = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1 \quad \lambda_1 = i, \lambda_2 = -i$$

Example 21.5

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 1 \\ 0 & 3-\lambda \end{vmatrix} = (3-\lambda)(3-\lambda)$$

$$\lambda_1 = 3, \lambda_2 = 3$$

$$(A - \lambda I)x = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x = 0 \quad x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} ? \\ ? \end{bmatrix}$$

No 2nd

independent

x_2

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Lecture 22. Diagonalization and Powers of A.

- Diagonalizing a matrix $S^{-1}AS = \Lambda$
- Powers of A / equation $u_{k+1} = Au_k \rightarrow$ 递推关系

Suppose n independent eigenvectors of A . Put them in the column of S .

$$AS = A \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \dots & x_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \lambda_1 x_1 & \lambda_2 x_2 & \dots & \lambda_n x_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \dots & x_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

$$= S\Lambda.$$

$$\boxed{AS = S\Lambda}$$

Diagonal eigenvalue
matrix Λ

$$S^{-1}AS = S^{-1}S\Lambda = \Lambda.$$

$$\boxed{A = S\Lambda S^{-1}}$$

If $Ax = \lambda x$, ① $A^2x = \lambda Ax = \lambda^2 x$

② $A^2 = S\Lambda S^{-1}S\Lambda S^{-1} = S\Lambda^2 S^{-1}$

$$\rightarrow A^k = S\Lambda^k S^{-1}$$

Theorem. $A^k \rightarrow 0$ as $k \rightarrow \infty$, if all $|\lambda_i| < 1$.

A is sure to have n independent eigenvectors (and be diagonalizable) if all the λ 's are different. (no repeated eigenvalues)

Matlab: eig(rand(10,10)).

Repeated eigenvalues: may or may not have n independent eigenvectors.

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \quad \det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{vmatrix} \quad \lambda = 2, 2.$$

$$A - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Equation $u_{k+1} = Au_k$

Start with given vector u_0

$$u_1 = Au_0$$

$$u_2 = A^2 u_0$$

$\vdots$

$$\boxed{u_k = A^k u_0}$$

To really solve: Write $u_0 = c_1 x_1 + c_2 x_2 + \dots + c_n x_n = SC$

$$Au_0 = c_1 \lambda_1 x_1 + c_2 \lambda_2 x_2 + \dots + c_n \lambda_n x_n = S \Lambda S^{-1} SC = S \Lambda C$$

$$\begin{aligned} A^{100} u_0 &= c_1 \lambda_1^{100} x_1 + c_2 \lambda_2^{100} x_2 + \dots + c_n \lambda_n^{100} x_n \\ &= S \Lambda^{100} C \end{aligned}$$

Example 22.1. Fibonacci: 0, 1, 1, 2, 3, 5, 8, 13, ..., F_{100} .

Find F_{100} .

$$\begin{aligned} F_{k+2} &= F_{k+1} + F_k. \quad \text{let } u_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} \Rightarrow u_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} u_k \\ F_{k+1} &= F_k + F_{k-1}. \end{aligned}$$

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad |A - \lambda I| = \begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = 0.$$

$$\begin{aligned} \lambda_1 &= \frac{1}{2}(1 + \sqrt{5}), \quad \lambda_2 = \frac{1}{2}(1 - \sqrt{5}) \\ &\approx 1.62, \quad \approx -0.62 \\ x_1 &= \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix}, \quad x_2 = \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \therefore F_{100} &= c_1 \left(\frac{1 + \sqrt{5}}{2} \right)^{100} + c_2 \left(\frac{1 - \sqrt{5}}{2} \right)^{100} \\ &\approx c_1 \left(\frac{1 + \sqrt{5}}{2} \right)^{100} \end{aligned}$$

$$u_0 = \begin{bmatrix} F_1 \\ F_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad c_1 x_1 + c_2 x_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{aligned} c_1 &= \\ c_2 &= \end{aligned}$$

$\rightarrow F_{100} \quad \checkmark$

Lecture 23. Differential Equations and $\exp(At)$

$$\begin{cases} \text{Differential Equations } \frac{du}{dt} = Au. \\ \text{Exponential } e^{At} \text{ of a matrix.} \end{cases}$$

Example 23.1.

$$\frac{du_1}{dt} = -u_1 + 2u_2 \quad u(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ (Initial Condition).}$$

$$\frac{du_2}{dt} = u_1 - 2u_2.$$

$$\rightarrow A = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix} \quad \lambda = 0, -3 \quad x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$\text{Solution: } u(t) = c_1 e^{\lambda_1 t} x_1 + c_2 e^{\lambda_2 t} x_2.$$

$$\text{CHECK: } \frac{du}{dt} = Au \quad \text{Plug in } e^{\lambda_1 t} x_1: \lambda_1 e^{\lambda_1 t} x_1 = A \cdot e^{\lambda_1 t} x_1. \quad \checkmark$$

$$= c_1 \cdot 1 \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \cdot e^{-3t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}.$$

$$\text{Use } u(0) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}: \text{ At } t=0 \leftrightarrow c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

$$\Rightarrow \begin{aligned} c_1 &= \frac{1}{3} \\ c_2 &= \frac{1}{3}. \end{aligned}$$

$$\downarrow \quad \underline{Sc = u(0)}.$$

$$\text{Steady State } u(\infty) = \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

$$\textcircled{1} \text{ Stability } u(t) \rightarrow 0. \quad \text{need } e^{\lambda t} \rightarrow 0 \quad \text{need } \operatorname{Re}[\lambda] < 0.$$

$$\textcircled{2} \text{ Steady State } \lambda_1 = 0 \text{ and other } \operatorname{Re}[\lambda] < 0.$$

$$\textcircled{3} \text{ Blow up if any } \operatorname{Re}[\lambda] > 0.$$

$$2 \times 2 \text{ Stability. } \operatorname{Re}[\lambda_1] < 0, \operatorname{Re}[\lambda_2] < 0.$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{Trace: } a+d = \lambda_1 + \lambda_2 < 0. \quad \begin{bmatrix} -2 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{Trace} = -1 \quad \text{But still blows up.}$$

$$\rightarrow \text{Additional condition: } \underline{\det > 0}.$$

$$\frac{du}{dt} = Au$$

$$\text{Set } u = Sv : S \frac{dv}{dt} = ASv$$

$$\frac{dv}{dt} = S^{-1}ASv$$

$$\frac{dv}{dt} = \Lambda v$$

$$\Rightarrow \frac{dv_i}{dt} = \lambda_i v_i$$

$$\Rightarrow v(t) = e^{\Lambda t} v(0)$$

$$u(t) = \underbrace{S e^{\Lambda t} S^{-1}}_{e^{At}} u(0) = e^{At} u(0)$$

Matrix exponential e^{At}

WHY?

$$e^{At} = I + At + \frac{(At)^2}{2} + \frac{(At)^3}{6} + \dots + \frac{(At)^n}{n!} + \dots$$

$$e^x = \sum \frac{x^n}{n!}$$

$$\left((I - At)^{-1} = I + At + (At)^2 + (At)^3 + \dots + (At)^n + \dots \quad \frac{1}{1-x} = \sum x^n \right)$$

$$= I + S \Lambda S^{-1} t + \frac{S \Lambda^2 S^{-1}}{2} t^2 + \dots + \frac{S \Lambda^n S^{-1}}{n!} t^n + \dots$$

$$= S \left(\Lambda t + \frac{\Lambda^2}{2} t^2 + \dots + \frac{\Lambda^n}{n!} t^n + \dots \right) S^{-1}$$

$$= S \underbrace{e^{\Lambda t}}_{e^{\Lambda t}} S^{-1}$$

$$e^{\Lambda t} : \Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ 0 & \lambda_2 & \\ & & \ddots \\ 0 & & & \lambda_n \end{bmatrix}$$

$$\Rightarrow e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & 0 \\ & e^{\lambda_2 t} & \\ 0 & & \ddots \\ & & & e^{\lambda_n t} \end{bmatrix} \quad \text{Re}(\lambda_i) < 0$$

Example 23.2

$$y'' + by' + ky = 0$$

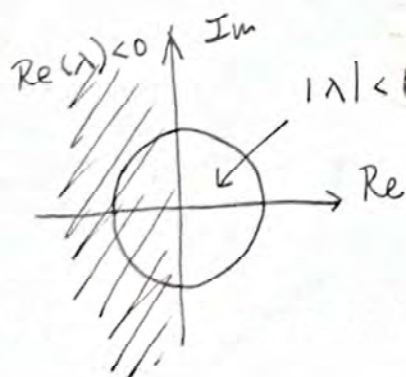
1st order equation

$\rightarrow 2 \times 2$ 1st order equation.

$$\rightarrow \text{Let } u = \begin{bmatrix} y' \\ y \end{bmatrix}, u' = \begin{bmatrix} y'' \\ y' \end{bmatrix} = \begin{bmatrix} -b & -k \\ 1 & 0 \end{bmatrix} \begin{bmatrix} y' \\ y \end{bmatrix}$$

$\rightarrow 5^{\text{th}}$ equation $\rightarrow 5 \times 5$ 1st order equation

$$\begin{bmatrix} 1 & - & - & - & - \\ & 1 & & & \\ & & 1 & & \\ & & & 1 & \\ & & & & 0 \end{bmatrix}$$



MIT 18.06 Linear Algebra

Lecture 24. Markov Matrices ; Fourier Series.

Markov matrices Steady State : $\lambda = 1$ Fourier Series + Projections

Example 24.1.

$$A = \begin{bmatrix} 0.1 & 0.01 & 0.3 \\ 0.2 & 0.99 & 0.3 \\ 0.7 & 0 & 0.4 \end{bmatrix}$$

Properties:

① all entries ≥ 0 .

② all columns add to 1.

1. $\lambda = 1$ is an eigenvalue
2. All other eigenvalues $|\lambda_i| < 1$.

$$u_k = A^k u_0 = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2 + \dots$$

$(\lambda_1 = 1) \quad \rightarrow 0$
 $\rightarrow c_1 x_1$
 $\uparrow$

x_1 part of u_0

 Steady State.

$$A - I = \begin{bmatrix} -0.9 & 0.01 & 0.3 \\ 0.2 & -0.01 & 0.3 \\ 0.7 & 0 & -0.6 \end{bmatrix}$$

All cols of $(A - I)$ add to zero $\rightarrow (A - I)$ is singular.

$\rightarrow$ because rows are dependent.

$\rightarrow$ because $(1, 1, 1)$ is in $N((A - I)^T)$.

then x_1 is in $N(A - I)$.

eigenvalues of A are the same as eigenvalues of A^T . $\left[\begin{array}{l} \det(A - \lambda I) = 0 \\ \det(A^T - \lambda I) = 0 \end{array} \right]$

$$\Rightarrow x_1 = \begin{bmatrix} 0.6 \\ 33 \\ 0.7 \end{bmatrix}$$

$$\begin{bmatrix} u_{cal} \\ u_{mass} \end{bmatrix}_{t=k+1} = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix} \begin{bmatrix} u_{cal} \\ u_{mass} \end{bmatrix}_{t=k}$$

$$\begin{bmatrix} u_{cal} \\ u_{mass} \end{bmatrix}_{t=0} = \begin{bmatrix} 0 \\ 1000 \end{bmatrix}$$

Applications of Markov Matrix.

$$\begin{bmatrix} u_{cal} \\ u_{mass} \end{bmatrix}_{t=1} = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix} \begin{bmatrix} 0 \\ 1000 \end{bmatrix} = \begin{bmatrix} 200 \\ 800 \end{bmatrix}$$

$\vdots$

Some eigenvalues and eigenvectors!

Example 24.2.

$u_{k+1} = A u_k$. A is Markov.

$$\begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}, \lambda_1 = 1, \lambda_2 = 0.7.$$

$$x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$u_k = c_1 \cdot 1^k \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \cdot 0.7^k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$u_0 = \begin{bmatrix} 0 \\ 1000 \end{bmatrix} = c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \cdot \begin{cases} c_1 = \frac{1000}{3} \\ c_2 = \frac{2000}{3} \end{cases}$$

Projection with orthonormal basis $q_1, q_2, \dots, q_n$.

expansion.

$$\text{Any } v = x_1 q_1 + x_2 q_2 + \dots + x_n q_n$$

$$q_1^T v = x_1 \underbrace{q_1^T q_1}_1 + 0 + \dots + 0 = x_1 \rightarrow \begin{bmatrix} q_1 & \dots & q_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = v$$

Fourier Series: $f(x) = f(x + 2\pi)$ Periodic functions.

$$f(x) = a_0 + a_1 \cos x + b_1 \sin x + a_2 \cos 2x + b_2 \sin 2x + \dots$$

$$\text{For vectors: } v^T w = v_1 w_1 + \dots + v_n w_n$$

$$\text{For Functions: } f^T g = \int_0^{2\pi} f(x) g(x) dx$$

$$\int_0^{2\pi} \sin x \cos x dx = \frac{1}{2} (\sin x)^2 \Big|_0^{2\pi} = 0. \quad - \checkmark$$

$$\int_0^{2\pi} f(x) \cos x dx = \int_0^{2\pi} a_1 (\cos x)^2 dx = \pi a_1$$

$$\Rightarrow \underline{a_1 = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos x dx}$$

MIT 18.06 Linear Algebra

Lecture 25. Symmetric Matrices and Positive Definiteness

Symmetric matrices
Eigenvalues / Eigenvectors

Start: Positive Definite Matrices

$$A = A^T.$$

① The eigenvalues are REAL.

② The eigenvectors are PERPENDICULAR.

Can be
chosen
(repeated
eigenvalue
case).

orthonormal eigenvectors.
columns of Q .

Usual: $A = S \Lambda S^{-1}$

Symmetric: $A = Q \Lambda Q^{-1} = Q \Lambda Q^T$

$Q^{-1} = Q^T$
for orthonormal matrix.

$$\begin{aligned} \bar{x}^T x &= [\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n] \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \\ &= \bar{x}_1 x_1 + \bar{x}_2 x_2 + \dots + \bar{x}_n x_n \\ &= \|x_1\|^2 + \|x_2\|^2 + \dots + \|x_n\|^2 > 0 \end{aligned}$$

Why real eigenvalues?

$$Ax = \lambda x \xrightarrow{\text{always}} A\bar{x} = \bar{\lambda} \bar{x} \Rightarrow \bar{x}^T A^T = \bar{x}^T \bar{\lambda}$$

$\Downarrow$

$$\bar{x}^T Ax = \lambda \bar{x}^T x$$

if $A^T = A$

$$\Rightarrow \bar{x}^T A = \bar{x}^T \bar{\lambda}$$

$\Downarrow$

$$\lambda \bar{x}^T x = \bar{\lambda} \bar{x}^T x \quad \bar{x}^T Ax = \bar{x}^T \bar{\lambda} x$$

$\Downarrow$

$$\lambda = \bar{\lambda} \rightarrow \boxed{\lambda \text{ is real.}}$$

Good Matrices

- Real λ 's
- Perp. x 's.

$A = A^T$ if A is real.

$A = \bar{A}^T$ if A is complex.

$$A = A^T.$$

Every symmetric matrix is IN a combination of perp. projection matrices.

$$A = Q \Lambda Q^T = \begin{bmatrix} q_1 & q_2 & \dots & q_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \dots & \\ & & & \lambda_n \end{bmatrix} \begin{bmatrix} q_1^T \\ q_2^T \\ \vdots \\ q_n^T \end{bmatrix}$$

$$= \lambda_1 \underbrace{q_1 q_1^T + \lambda_2 q_2 q_2^T + \dots + \lambda_n q_n q_n^T}_{\text{projection matrix.}}$$

Signs of pivots are same as signs of λ 's.

$$\begin{cases} \# \text{ positive pivots} = \# \text{ positive } \lambda \text{'s.} \\ \# \text{ negative pivots} = \# \text{ negative } \lambda \text{'s.} \end{cases}$$

Positive Definite Symmetric Matrix.

all eigenvalues are positive.

all pivots are positive.

all sub-determinants are positive.



Example 25.1

$$\begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix} \quad \text{pivots: } 5, \frac{11}{5}$$

$$\lambda = 4 \pm \sqrt{5}$$

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Lecture 26. Complex Matrices ; FFT

Complex [vectors
matrices Fourier Matrix F_n .
inner products
Discrete Fourier Transform
Fast Fourier Transform n^2 mults $\rightarrow n \log n$

$\vec{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{bmatrix}$ in $\mathbb{C}^N$. length: $\sqrt{\vec{z}^T \vec{z}}$ is no good. $\sqrt{\vec{z}^H \vec{z}}$ is good. $\Rightarrow \sqrt{\vec{z}^H \vec{z}}$ Hermite

Example 26.1 length of $\begin{bmatrix} 1 \\ i \end{bmatrix}$: $\sqrt{\begin{bmatrix} 1 & -i \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix}} = \sqrt{1+1} = \sqrt{2}$.

Inner Product: $\vec{y}^T \vec{x} = \vec{y}^H \vec{x}$.

Example: $\begin{bmatrix} 2 & 3+i \\ 3-i & 5 \end{bmatrix}$.

Symmetric: $A^T = A$ no good. if A is complex.
 $\downarrow$
Hermitian: $\overline{A}^T = A$ good.
Matrix $A^H = A$

Perpendicular: $q_1, q_2, \dots, q_n$. $q_i^H q_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$
 $Q = [q_1 \ q_2 \ \dots \ q_n]$ $Q^H Q = I$ orthogonal $\rightarrow$ unitary

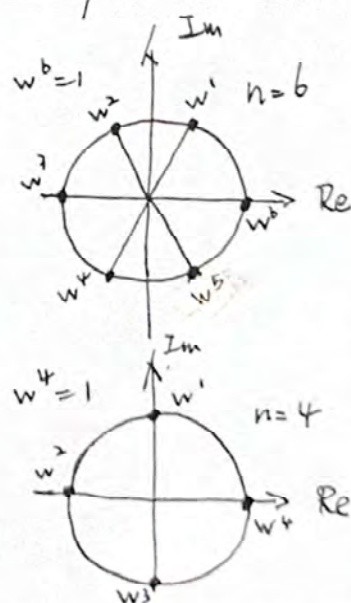
Fourier Matrix.

$$F_n = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & w & w^2 & \dots & w^{n-1} \\ 1 & w^2 & w^4 & \dots & w^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & w^{n-1} & w^{2(n-1)} & \dots & w^{(n-1)^2} \end{bmatrix}_{n \times n}$$

$$w = e^{i \frac{2\pi}{n}} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$$

$$F_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & i & -1 & -i \end{bmatrix}$$

$$F_4^H F_4 = I.$$



$$W_{64}^2 = e^{i \frac{2\pi}{64} \cdot 2} = e^{i \frac{2\pi}{32}} = W_{32}$$

$$[F_{64}] = \begin{bmatrix} I_{32} & D_{32} \\ I_{32} & -D_{32} \end{bmatrix} \begin{bmatrix} F_{32} & 0 \\ 0 & F_{32} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{bmatrix}$$

$\downarrow$
 64^2

$2(32)^2 + 32$

$$= \begin{bmatrix} I_{32} & D_{32} \\ I_{32} & -D_{32} \end{bmatrix} \begin{bmatrix} I_{16} & D_{16} & 0 \\ I_{16} & -D_{16} & 0 \\ 0 & 0 & I_{16} & D_{16} \\ 0 & 0 & I_{16} & -D_{16} \end{bmatrix} \begin{bmatrix} F_{16} & 0 \\ 0 & F_{16} \\ 0 & F_{16} \\ 0 & F_{16} \end{bmatrix} \begin{bmatrix} P_{16} & 0 \\ 0 & P_{16} \\ 0 & P_{16} \\ 0 & P_{16} \end{bmatrix} \begin{bmatrix} P_{32} \\ \vdots \\ P_{32} \end{bmatrix}$$

$$\downarrow$$

$$2(2(16)^2 + 16) + 32$$

$$\downarrow$$

$$\log 64 \cdot \frac{64}{2}$$

= ...

$$\Downarrow$$

$\frac{n}{2} \log n$

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Lecture 27 Positive Definite Matrices and Minima.

- Positive Definite Matrix (Tests)
- Tests for minimum ($x^T A x > 0$)
- Ellipsoids in $\mathbb{R}^n$

$$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix} \quad \left. \begin{array}{l} \textcircled{1} \lambda_1 > 0, \lambda_2 > 0 \\ \textcircled{2} a > 0, ac - b^2 > 0 \\ \textcircled{3} \text{ pivots } a > 0, \frac{ac-b^2}{a} > 0 \\ * \textcircled{4} x^T A x > 0 \end{array} \right\} \text{positive definite}$$

Example 27.1

$$\begin{bmatrix} 2 & 6 \\ 6 & - \end{bmatrix} \quad ac - b^2 = 2c - 36 > 0 \quad c > 18.$$

if $c = 18$ - semidefinite: $\lambda_1 = 0, \lambda_2 = 20$. pivot: 2

positive

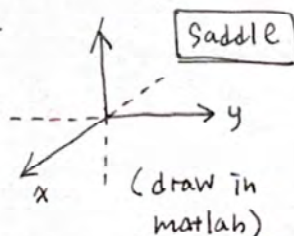
$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad x^T A x = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 2 & 6 \\ 6 & 18 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \underbrace{2x_1^2}_{ax^2} + \underbrace{12x_1x_2}_{2bxy} + \underbrace{18x_2^2}_{cy^2}$$

quadratic form

Graphs of $f(x,y) = x^T A x = ax^2 + bxy + cy^2$

$$\begin{bmatrix} 2 & 6 \\ 6 & 7 \end{bmatrix} \Rightarrow 2x^2 + 12xy + 7y^2$$

Non-positive definite

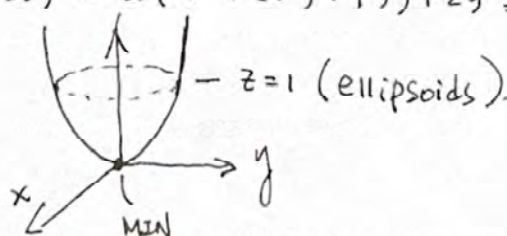


$$\begin{bmatrix} 2 & 6 \\ 6 & 20 \end{bmatrix} \Rightarrow 2x^2 + 12xy + 20y^2 = 2(x^2 + 6xy + 9y^2) + 2y^2 = 2(x+3y)^2 + 2y^2$$

pivots
multiplier

Positive definite

$$\begin{cases} 1^{st} \text{ deriv} = 0 \\ 2^{nd} \text{ deriv} > 0 \end{cases}$$



Calculus: $\frac{d^2 u}{dx^2} > 0$

Linear Algebra: Matrix of 2nd derivs.

$$\begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

Example 27.2 3x3 case.

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

det's: 2, 3, 4. > 0

pivots: 2, $\frac{3}{2}$, $\frac{4}{3}$ > 0

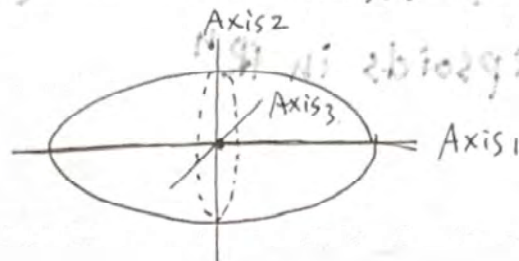
λ 's: $2 - \sqrt{2}$, 2, $2 + \sqrt{2}$ > 0

$x^T A x = 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3 > 0$

if $x^T A x = 1$:

$$A = Q \Lambda Q^T$$

The principle Axis.



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Lecture 28. Similar Matrices and Jordan Form

$$\left\{ \begin{array}{l} A^T A \text{ is positive definite!} \\ \text{Similar matrices } A, B \text{ / Jordan} \\ B = M^{-1} A M \text{ / Form} \end{array} \right.$$

positive definite means: $x^T A x > 0$ (except for $x = 0$)

• If A, B are positive definite, so is $A+B$.

$$x^T (A+B) x = x^T A x + x^T B x > 0$$

• If A is positive definite, so is A^{-1} .

$$A = S \Lambda S^{-1}, \quad A^{-1} = S \Lambda^{-1} S^{-1}$$

Now $A_{n \times n}$: $A^T A \rightarrow$ square, symmetric. Positive definite?

$$x^T A^T A x = (Ax)^T (Ax) = \|Ax\|^2 \geq 0.$$

$A_{n \times n}$ and $B_{n \times n}$ are similar. means for some M : $B = M^{-1} A M$.

Example 28.1: A is similar to Λ .

$$S^{-1} A S = \Lambda.$$

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad \Lambda = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}, \quad M^{-1} = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}$$

$$M^{-1} A M = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & -15 \\ 1 & 6 \end{bmatrix} = \underline{\underline{B}}.$$

Similar matrices have the same eigenvalues. (same # eigenvectors)

$$\rightarrow Ax = \lambda x$$

$$A M M^{-1} x = \lambda x$$

$$(M^{-1} A M) M^{-1} x = \lambda M^{-1} x$$

$$B(M^{-1} x) = \lambda(M^{-1} x)$$

λ

Eigenvector of $B = M^{-1}$ (eigenvector of A)

Bad cases, $\lambda_1 = \lambda_2 = 4$

one family has $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} \Rightarrow M^{-1} \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} M = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$ limit? 8.5 similar.

The big family $\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix}$
Includes:

Jordan Form

More members of family:

$$\begin{bmatrix} 5 & 1 \\ -1 & 3 \end{bmatrix}, \begin{bmatrix} 4 & 0 \\ 1 & 4 \end{bmatrix}, \dots, \begin{bmatrix} a & * \\ * & 8-a \end{bmatrix} \begin{matrix} \text{Trace} = 8. \\ \text{det} = 16. \end{matrix}$$

Bad Case 2.

$$\lambda = 0, 0, 0, 0$$

$$\left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

2 eigenvectors, 2 missing.

$$\dim N(A) = 2.$$

$$\left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

2 eigenvectors, 2 missing.

$$\dim N(A) = 2.$$

Jordan Block / 1 eigenvector.

$$J_i = \begin{bmatrix} \lambda_i & 1 & 0 \\ 0 & \lambda_i & 1 \\ & & \ddots & \lambda_i \end{bmatrix}$$

Not similar because the size of Jordan Block is not the same.

• Every square A is similar to a Jordan matrix J .

$$J = \begin{bmatrix} \boxed{J_1} & & \\ & \boxed{J_2} & \\ & & \ddots \\ & & & \boxed{J_d} \end{bmatrix} \quad \# \text{ blocks} = \# \text{ Eigenvalues}$$

Good case: $J = \lambda$.

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Lecture 29. Singular Value Decomposition

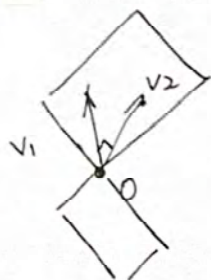
$$\begin{cases} \text{Singular Value Decomposition} = \text{SVD} \\ A = U \Sigma V^T \quad // \quad \Sigma \text{ diagonal} \\ \quad \quad \quad \quad \quad \quad \quad u, v, \text{ orthogonal.} \end{cases}$$

Symmetric positive definite

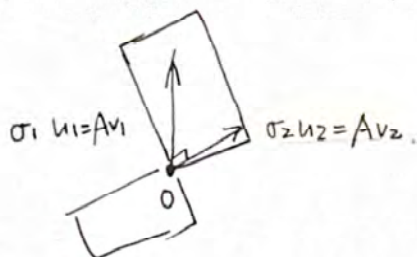
$$A = Q \Lambda Q^T \quad \checkmark$$

$$A = S \Lambda S^{-1} \quad \times \quad S \text{ is not orthogonal.}$$

$\mathbb{R}^n$ Row Space



Column Space $\mathbb{R}^m$



orthogonal $v_1, \dots, v_r$.

$$A \begin{bmatrix} v_1 & v_2 & \dots & v_r \end{bmatrix} = \begin{bmatrix} u_1 & u_2 & \dots & u_r \end{bmatrix} \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{bmatrix}$$

($N(A), N(A^T)$ included.)

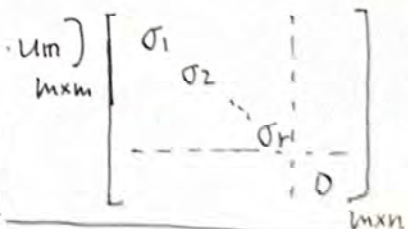
$$\Rightarrow AV = U\Sigma$$

Example 29.1

$$A = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix}$$

v_1, v_2 in row space $\mathbb{R}^2$
 u_1, u_2 in column space $\mathbb{R}^2$
 $\sigma_1 > 0, \sigma_2 > 0$

$$\begin{cases} Av_1 = \sigma_1 u_1 \\ Av_2 = \sigma_2 u_2 \end{cases}$$



$$A = U \Sigma V^{-1} = U \Sigma V^T$$

$$A^T A = V \Sigma^T \underline{U^T U} \Sigma V^T = V \begin{bmatrix} \sigma_1^2 & & \\ & \sigma_2^2 & \\ & & \ddots \\ & & & \sigma_n^2 \end{bmatrix} V^T \quad \text{--- } Q \Lambda Q^T \text{ Form}$$

Example 29.1 Solution.

$$A^T A = \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 25 & 7 \\ 7 & 25 \end{bmatrix}$$

$$\lambda_1 = 32, \sigma_1 = \sqrt{32}, \lambda_2 = 18, \sigma_2 = \sqrt{18}$$

$$x_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \sqrt{32} & 0 \\ 0 & \sqrt{18} \end{bmatrix} \quad V^T = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

→ Find u 's: $AA^T = U \Sigma V^T V \Sigma^T U^T = U \Sigma \Sigma^T U^T$

$$AA^T = \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} \begin{bmatrix} 4 & -3 \\ 4 & 3 \end{bmatrix} = \begin{bmatrix} 32 & 0 \\ 0 & 18 \end{bmatrix}$$

$$\lambda_1 = 32, \sigma_1 = \sqrt{32}$$

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\lambda_2 = 18, \sigma_2 = \sqrt{18}$$

$$x_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} 4 & 4 \\ -3 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \sqrt{32} & 0 \\ 0 & \sqrt{18} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}$$

$$A = U \Sigma V^T$$

$$\boxed{AV = U\Sigma}$$

V is determined.

So U should satisfy the condition.

So no $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

Example 29.2 Singular.

$$A = \begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix}$$

row space: multiples of $\begin{bmatrix} 4 \\ 3 \end{bmatrix}$
nullspace

column space: multiples of $\begin{bmatrix} 4 \\ 8 \end{bmatrix}$
left null space

$$v_1 = \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}, u_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} \sqrt{125} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0.8 & 0.6 \\ 0.6 & -0.8 \end{bmatrix}$$

$$A = U \Sigma V^T$$

$v_1, \dots, v_r$ - Orthonormal basis for row space
 $u_1, \dots, u_r$ - Orthonormal basis for column space
 $v_{r+1}, \dots, v_n$ - Orthonormal basis for nullspace
 $u_{r+1}, \dots, u_m$ - Orthonormal basis for left nullspace.

and

$$Av_i = \sigma_i u_i$$

MIT 18.06 Linear Algebra

Lecture 30. Linear Transformations and Their Matrices

Linear Transformations T
without coordinates: no matrix
with coordinates $\mapsto$ Matrix

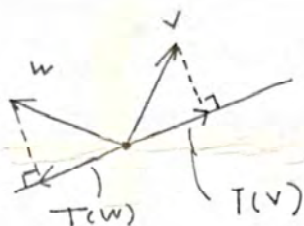
$$T(v+w) = T(v) + T(w).$$

$$T(cv) = cT(v)$$

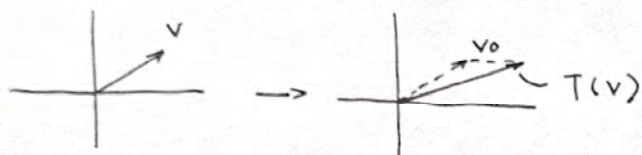
$$T(cv+dw) = cT(v) + dT(w)$$

Example 30.1. Projection.

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$



Example 30.2. Shift the whole plane by v_0 . $\Rightarrow$ not a linear transformation.

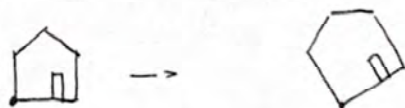
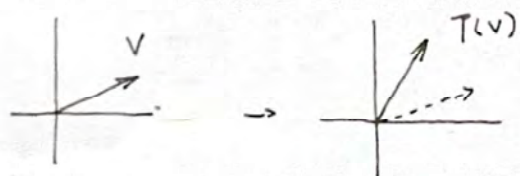


Example 30.3. $\rightarrow$ not a linear transformation.

$$T(v) = \|v\| \quad T: \mathbb{R}^3 \rightarrow \mathbb{R}^1$$

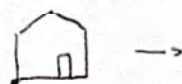
Example 30.4 Rotation by 45° .

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$



Example 30.5 Matrix A .

$$T(v) = Av. \quad A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$



Start: $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$

Example: $T(v) = Av, A_{2 \times 3}$

Information needed to know $T(v)$ for all inputs

$T(v_1), T(v_2), \dots, T(v_n)$ for all basis $v_1, v_2, \dots, v_n$.

Every $v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$.

Know $T(v) = c_1 T(v_1) + c_2 T(v_2) + \dots + c_n T(v_n)$

Coordinates come from a basis.

" of $v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n$

$$v = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Construct matrix A that represents linear transformation T . ($T: \mathbb{R}^n \rightarrow \mathbb{R}^m$)

Choose $v_1, v_2, \dots, v_n$ for input $\mathbb{R}^n$

Choose $w_1, w_2, \dots, w_m$ for output $\mathbb{R}^m$

Want A .

Eigenvector Basis leads to diagonal matrix Λ .

Example 30.6

$v = c_1 v_1 + c_2 v_2$
 $T(v) = c_1 v_1$
 $\Rightarrow (c_1, c_2) \rightarrow (c_1, 0)$
 $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$

$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} c_1 \\ 0 \end{bmatrix}$$

$\downarrow$ $\downarrow$ $\downarrow$
 A Input output

Proj onto 45° line. use standard $\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.
 $v_1=w_1$ $v_2=w_2$

$P = \frac{aa^T}{a^T a} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$ not so good. ? Why Column?

Example 30.7 $T = \frac{d}{dx}$

Input: $c_1 + c_2 x + c_3 x^2$

Output: $c_2 + 2c_3 x$

basis $\begin{cases} \text{In: } 1, x, x^2 \\ \text{Out: } 1, x \end{cases}$

$$A \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} c_2 \\ 2c_3 \end{bmatrix}$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}_{2 \times 3}$$

Rule to find A . Given the basis $v_1, \dots, v_n$ $w_1, \dots, w_m$

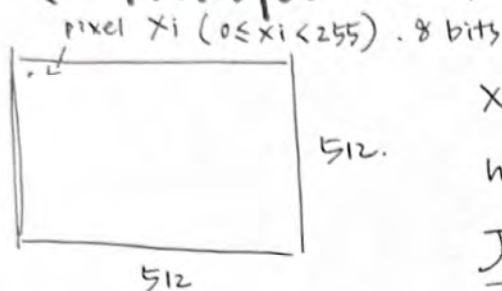
1st column of A : Write $T(v_1) = a_{11} w_1 + \dots + a_{m1} w_m$

2nd column of A : $T(v_2) = a_{12} w_1 + \dots + a_{m2} w_m$

$\vdots$
 $\downarrow$
 $A \begin{pmatrix} \text{input} \\ \text{coords} \end{pmatrix} = \begin{pmatrix} \text{output} \\ \text{coords} \end{pmatrix}$

Lecture 31. Change of Basis; Image Compression

Change of Basis
 Compression of Images
 Transformation $\leftrightarrow$ Matrix.



$$X \in \mathbb{R}^n$$

$$n = (512)^2$$

JPEG - Change of Basis. - Fourier basis, 8×8 .

Standard Basis:

$$\begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$$

Better Basis:

$$\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ \vdots \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ \vdots \\ -1 \\ \vdots \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ w_4 \\ w_5 \\ w_6 \\ w_7 \end{bmatrix}, \dots$$

Good Basis:

1. Fast. (FFT) (FWT)
 $\hookrightarrow$ orthogonal. $w^{-1} = w^T$.
2. Few is enough.

Signal P

$\downarrow$ change basis (lossless)

coefficients C

Linear Algebra does.

compression (lossy)

coefficients $\hat{C}$ & (many zeros)

$$\hat{X} = \sum \hat{C}_i v_i$$

$\uparrow$ reconstruct.

Video - Sequence of images.

correlated.

Wavelets 8×8

$$\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ \vdots \\ -1 \\ \vdots \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ \vdots \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ \vdots \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

In standard basis:

$$P = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \\ p_8 \end{bmatrix} = c_1 w_1 + \dots + c_8 w_8 = [W] \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_8 \end{bmatrix}$$

$$P = WC$$

$$C = W^{-1}P$$

Change of basis.

Columns of W = new basis vectors.

$$[x]_{\text{old basis}} \rightarrow [c]_{\text{new basis.}}$$

$$x = Wc$$

T with respect to $v_1, \dots, v_8$, it has matrix A .
with respect to $u_1, \dots, u_8$, it has matrix B .

relation: similar.

$$B = M^{-1} A M.$$

What is A ? using basis $v_1, \dots, v_8$.

Know T completely, from $T(v_1), T(v_2), T(v_3), \dots, T(v_8)$.

Because every $x = c_1 v_1 + c_2 v_2 + \dots + c_8 v_8$.

$$\text{Then } T(x) = c_1 T(v_1) + c_2 T(v_2) + \dots + c_8 T(v_8).$$

$$\text{Write } T(v_1) = a_{11} v_1 + a_{21} v_2 + \dots + a_{81} v_8.$$

$$T(v_2) = a_{12} v_1 + a_{22} v_2 + \dots + a_{82} v_8.$$

$\vdots$

$$[A] = \begin{bmatrix} a_{11} & \dots & a_{18} \\ \vdots & \ddots & \vdots \\ a_{81} & \dots & a_{88} \end{bmatrix}$$

Eigenvector basis $T(v_i) = \lambda_i v_i$. What is A ?

1st input is v_1 . its output is $\lambda_1 v_1$.

2nd input is v_2 . its output is $\lambda_2 v_2$.

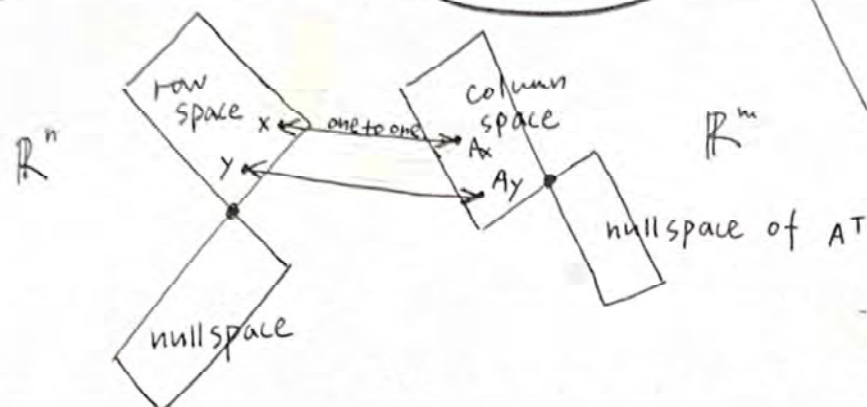
$\vdots$

$$A = \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \\ & & & \lambda_n \end{bmatrix}$$

Lecture 33. Left and Right Inverses ; Pseudo-inverse.

4 subspaces
 Left-inverses
 Right-inverses

Pseudo-inverses



$(x \neq y)$.
 If x, y in row space,
 then $Ax \neq Ay$.

$$x \xrightarrow[A^+]{A} Ax$$

$$y \xrightarrow[A^+]{A} Ay$$

$\rightarrow$ suppose $Ax = Ay$.

$$A(x - y) = 0.$$

in $N(A)$

Also in row space.

=

~~X~~

① 2-Sided inverse : $AA^{-1} = I = A^{-1}A$. ($r = m = n$)
 Full rank.

② Left-inverse : $A_{\text{left}}^{-1} = (A^T A)^{-1} A^T$

$$\rightarrow A_{\text{left}}^{-1} A = (A^T A)^{-1} A^T A = I_{n \times n}$$

$$\rightarrow A A_{\text{left}}^{-1} = A (A^T A)^{-1} A^T = P.$$

($r = n < m$)

Full column rank.

Nullspace = $\{0\}$

0 or 1 solutions to $Ax = b$.

③ Right-inverse : $A_{\text{right}}^{-1} = A^T (A A^T)^{-1}$

$$\rightarrow A A_{\text{right}}^{-1} = A A^T (A A^T)^{-1} = I_{m \times m}$$

$$\rightarrow A_{\text{right}}^{-1} A = A^T (A A^T)^{-1} A = P'$$

(onto row space).

($r = m < n$)

Full row rank.

$N(A^T) = \{0\}$

∞ solutions to $Ax = b$.

④ Pseudo inverse : $A^+ = V \Sigma^+ U^T$.

Find the pseudo inverse A^+

① Start from SVD : $A = U \Sigma V^T$

$$\Sigma^+ = \begin{bmatrix} 1/\sigma_1 & & & 0 \\ & 1/\sigma_2 & & \\ & & \ddots & \\ 0 & & & 1/\sigma_r & & 0 \\ & 0 & & & \ddots & \\ & & & & & 0 \end{bmatrix}_{n \times m}$$

$$\textcircled{2} A^+ = V \Sigma^+ U^T$$

Mit 18.08. 1908

an die Herren: Herrn Dr. v. ...
Herrn Dr. v. ...
Herrn Dr. v. ...

Sehr geehrte Herren!

Ich habe die Ehre, Ihnen

hiermit zu übersenden

mit der Bitte um

Erstattung